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Developing Robust and Interoperable Information Systems for SSH Research

The LOD4HSS Initiative

OBELIS

Summer School on Elites, Power and Inequality
17 June 2026

1.

New opportunities and challenges in the digital era

for research in

the humanities and social sciences (HSS)

Deutsche Forschungsgemeinschaft
(German Research Foundation)

The Digital Turn in the Sciences and Humanities

White Paper 2020
https://doi.org/10.5281/zenodo.4191345

- emergence of *new,* ***digital research practices*** which need to be integrated, *epistemically* and otherwise, in the context of a ***given discipline***.
- what is actually decisive for the expansion of knowledge-building possibilities are the associated ***scaling effects*** (volumes, speed, complexity)

The effects of the digital turn on the sciences and humanities can be divided into three categories:

- **Transformative change** concerns the transfer of analogue information and practices to a digital form
- **Enabling change** is the use of data-intensive technologies to *address [new] research questions* that could not be tackled in another form
- ***Substitutive change***, digital technologies are used to support or even replace conceptual parts of the research process

DFG, White Paper 2020

The process of knowledge production

in HSS disciplines

and

the possible impact of research digitisation

The epistemological cycle in natural sciences

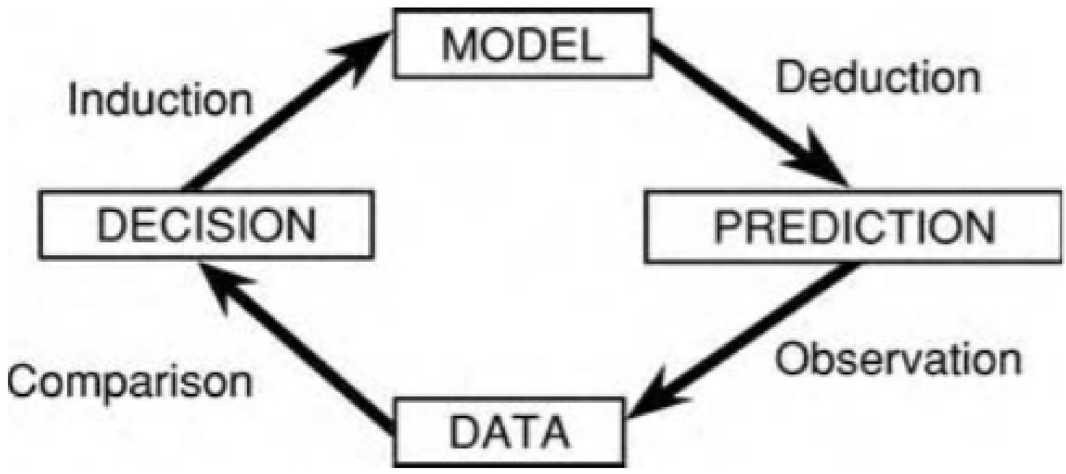
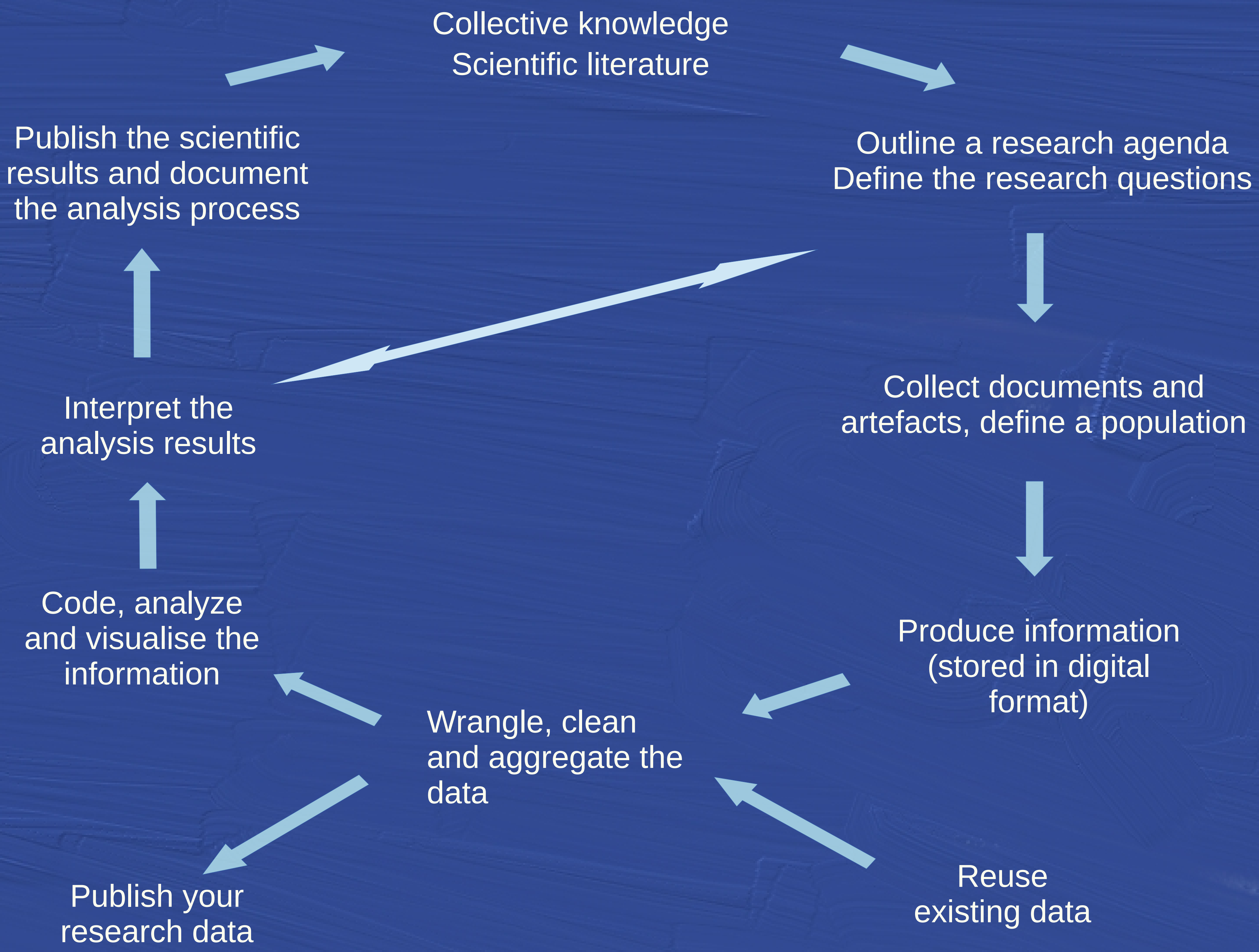
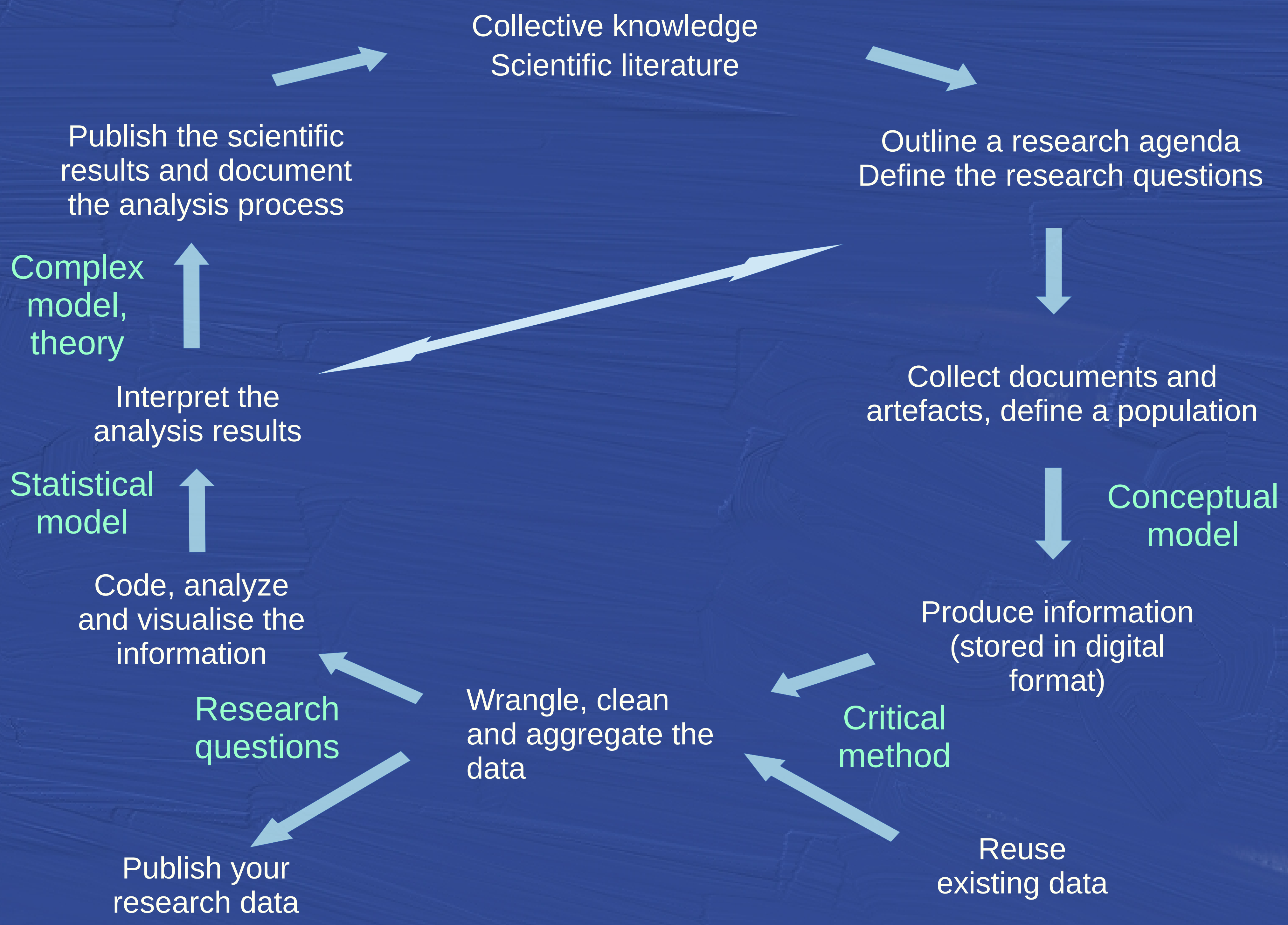
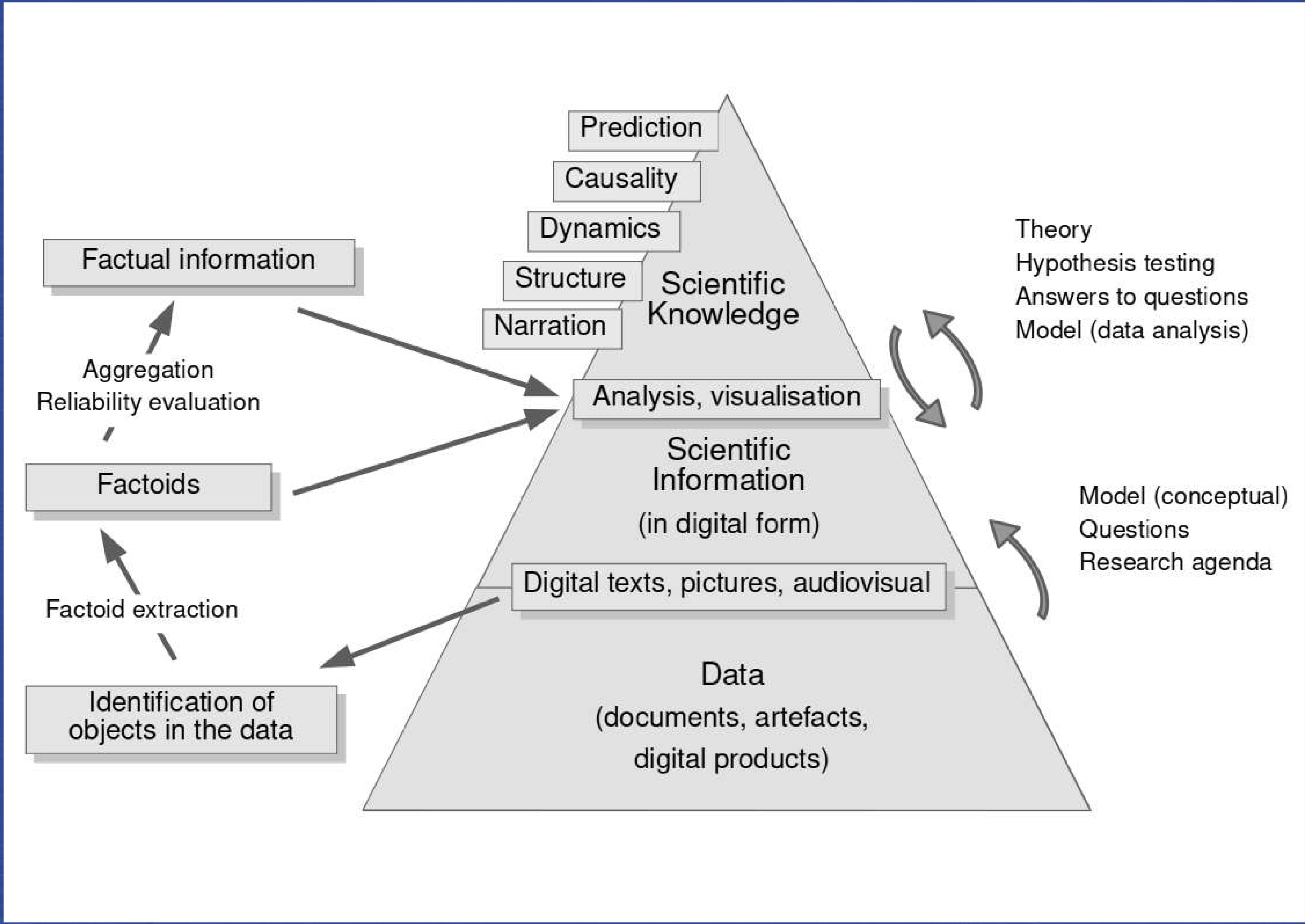


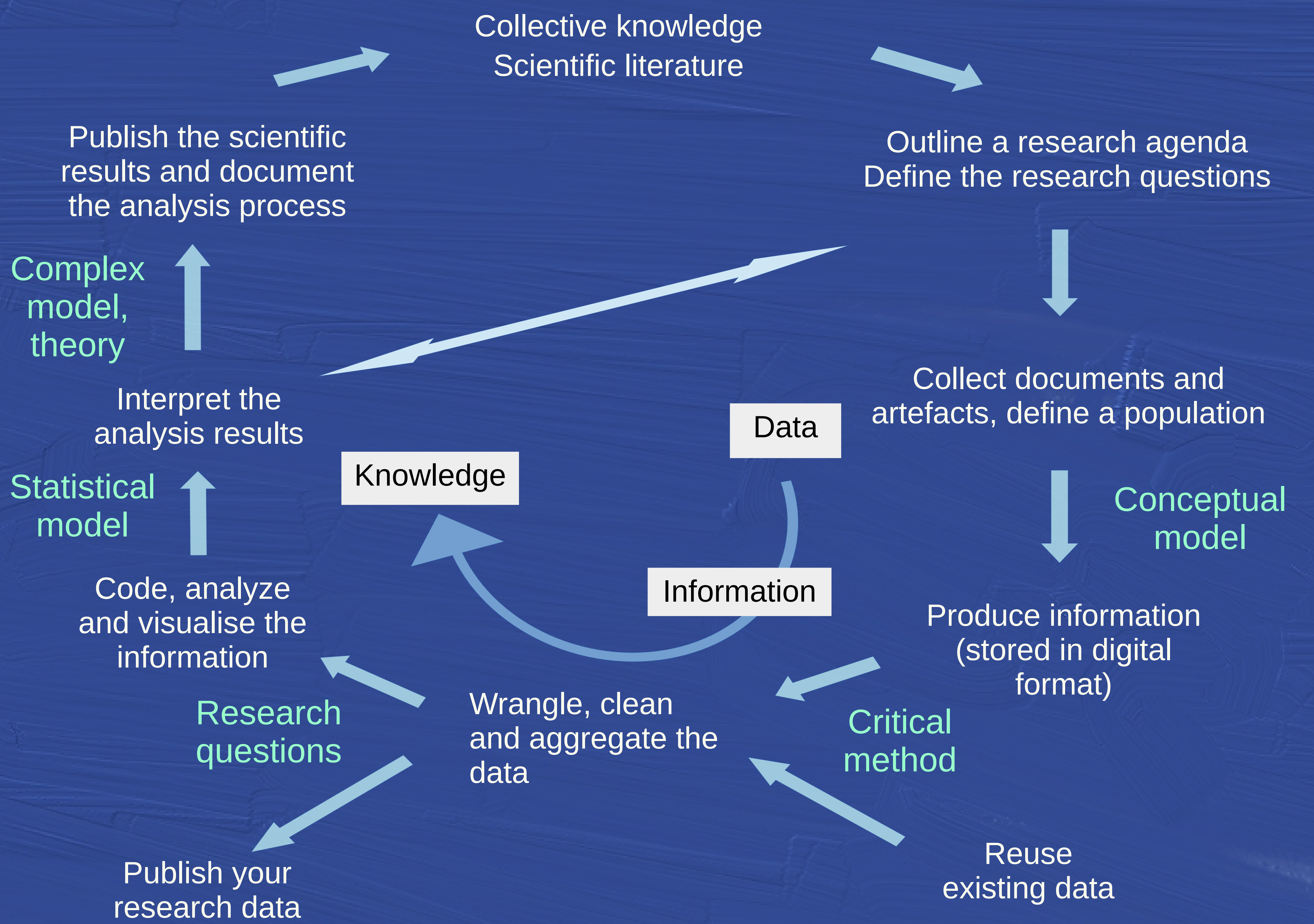
FIGURE 1.1 A schematic representation of the epistemological cycle

Hailman Jack P. / Strier Karen B., Planning, Proposing, and Presenting Science Effectively. A Guide for Graduate Students and Researchers in the Behavioral Sciences and Biology, 2nd ed., Cambridge, Cambridge University Press, 2006, 3.


Collective knowledge
Scientific literature
Outline a research agenda
Define the research questions
Collect documents and
artefacts, define a population
Produce information
(stored in digital
format)
Reuse
existing data
Wrangle, clean
and aggregate the
data
Publish your
research data
Code, analyze
and visualise the
information
Interpret the
analysis results
Publish the scientific
results and document
the analysis process


Collective knowledge
Scientific literature
Outline a research agenda
Define the research questions
Collect documents and artefacts, define a population
Conceptual model
Produce information (stored in digital format)
Critical method
Reuse existing data
Wrangle, clean and aggregate the data
Research questions
Publish your research data
Code, analyze and visualise the information
Statistical model
Interpret the analysis results
Complex model, theory
Publish the scientific results and document the analysis process


Prediction
Causality
Dynamics
Structure
Narration
Scientific Knowledge
Factual information
Aggregation
Reliability evaluation
Factoids
Factoid extraction
Identification of objects in the data
Analysis, visualisation
Scientific Information
(in digital form)
Digital texts, pictures, audiovisual
Data
(documents, artefacts, digital products)
Theory
Hypothesis testing
Answers to questions
Model (data analysis)
Model (conceptual)
Questions
Research agenda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Research questions
Code, analyze and visualise the information
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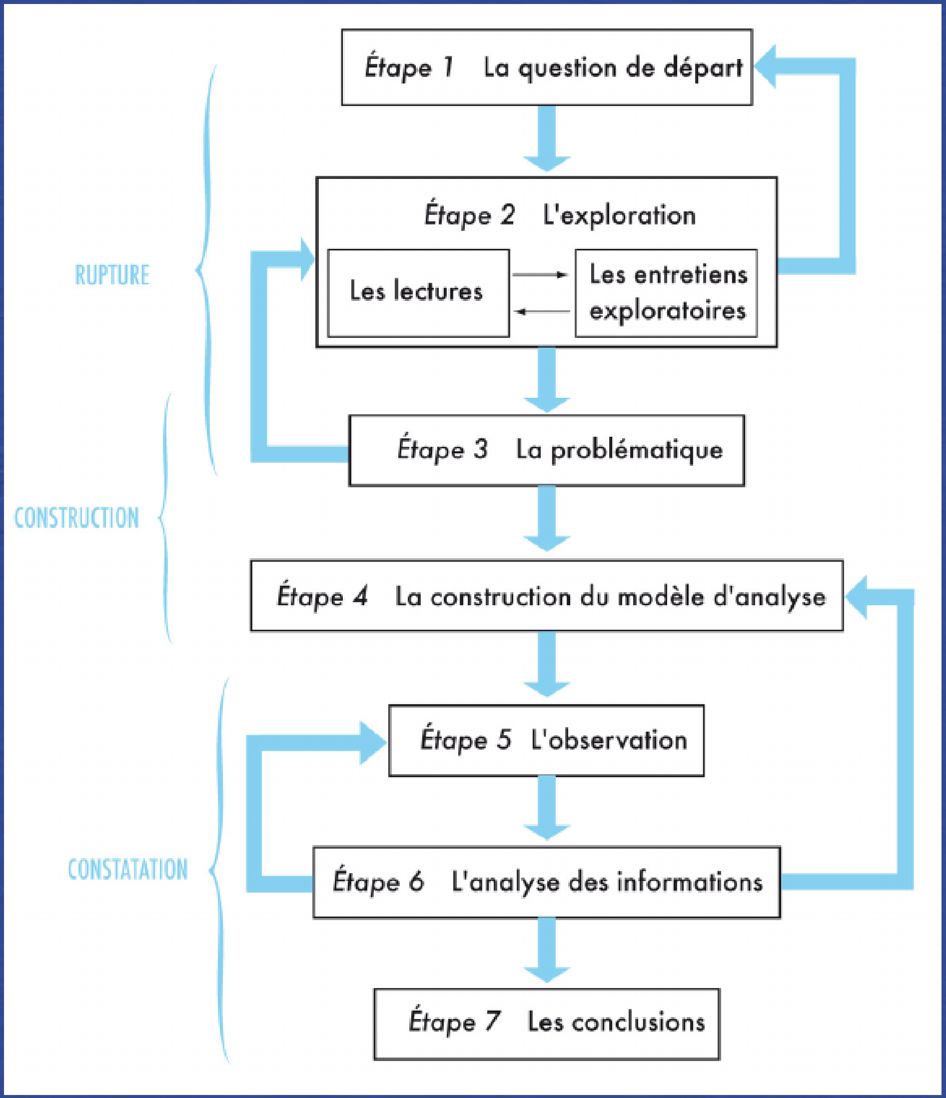
Interpret the analysis results
Complex model, theory
Publish the scientific results and document the analysis process
Data
Information
Knowledge



Van Campenhoudt Luc, Marquet Jacques et Quivy Raymond, *Manuel de recherche en sciences sociales*, 6e édition, Armand Colin, 2022.

Relevant aspects of the impact of digitisation on HSS

- Broadening the empirical basis of research results and models

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- New opportunities to develop research questions and theories

Relevant aspects of the impact of digitisation on HSS

- Broadening the empirical basis of research results and models
- New opportunities to develop research questions and theories
- Reproducibility and replicability of the research cycle based on *open research data* publication

2.

The challenge :

how can we make this transition and digital innovation agenda happen in HSS research ?

The method :

The Semantic Web and the FAIR principles

From spreadsheets / research databases

to Linked Open Data (LOD)

A blueprint : the LESSH Project

(follow this link)

Linked Open Data and the Semantic Web

Tim Berners-Lee, the inventor of the Web and Linked Data initiator, suggested a 5-star deployment scheme for Open Data.

https://5stardata.info/en/

★ make your stuff available **on the Web** (whatever format) under an open licence

★★ make it available as **structured data** (e.g., Excel instead of image scan of a table)

★★★ make it available in a non-proprietary **open format** (e.g., CSV instead of Excel)

★★★★ use **URIs to denote things**, so that people can point at your stuff

★★★★★ **link your data** to other data to provide context

Quoted from : https://5stardata.info/en/

Linked Open Data and the Semantic Web

(https://www.w3.org/TR/rdf11-concepts/)

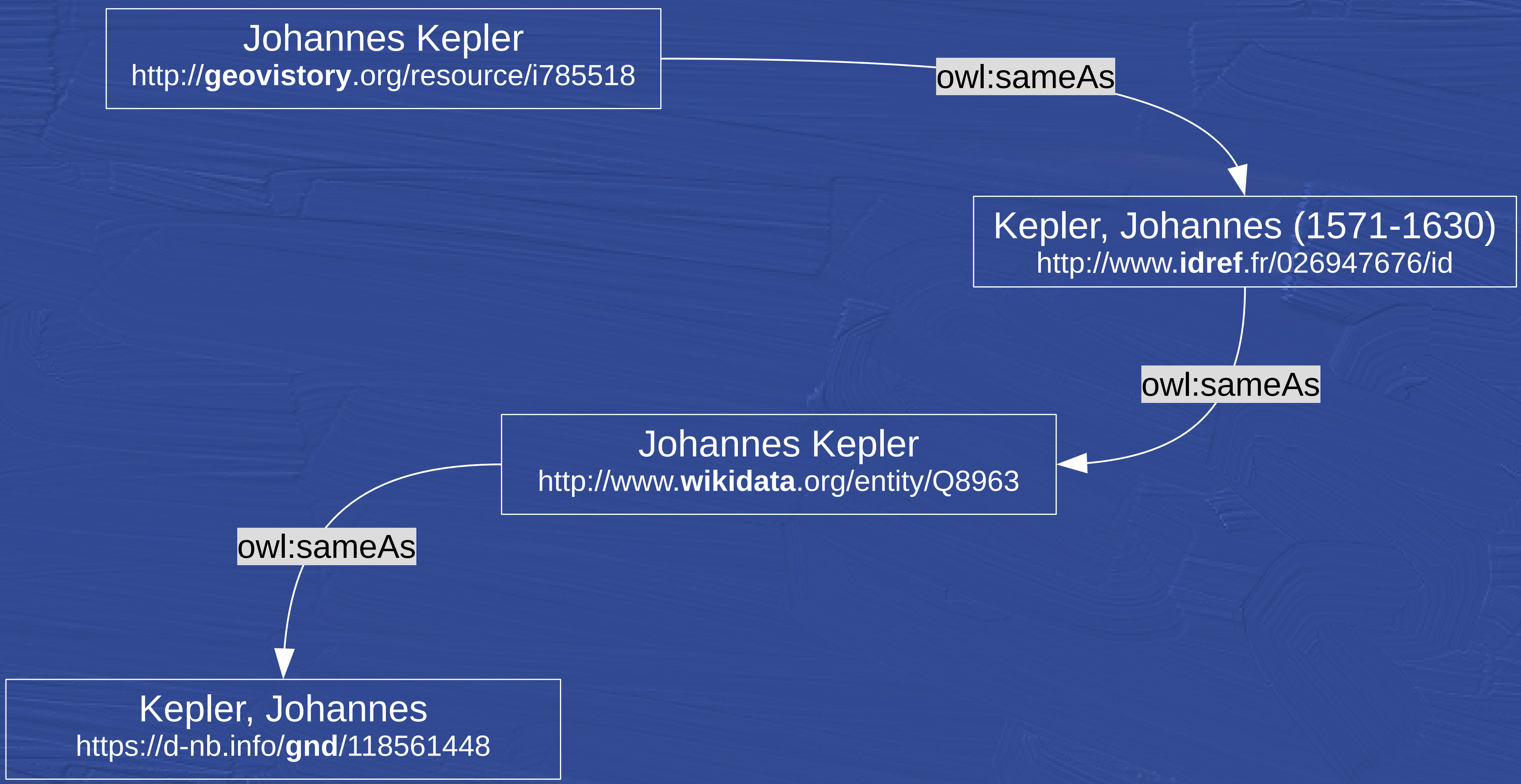
- « The Resource Description Framework (**RDF**) is a framework for **representing information in the Web**. »

- « A graph-based data model »

RDF
triple

LOD as Links

« URIs to denote things, so that people can point at your stuff »



LOD as Links

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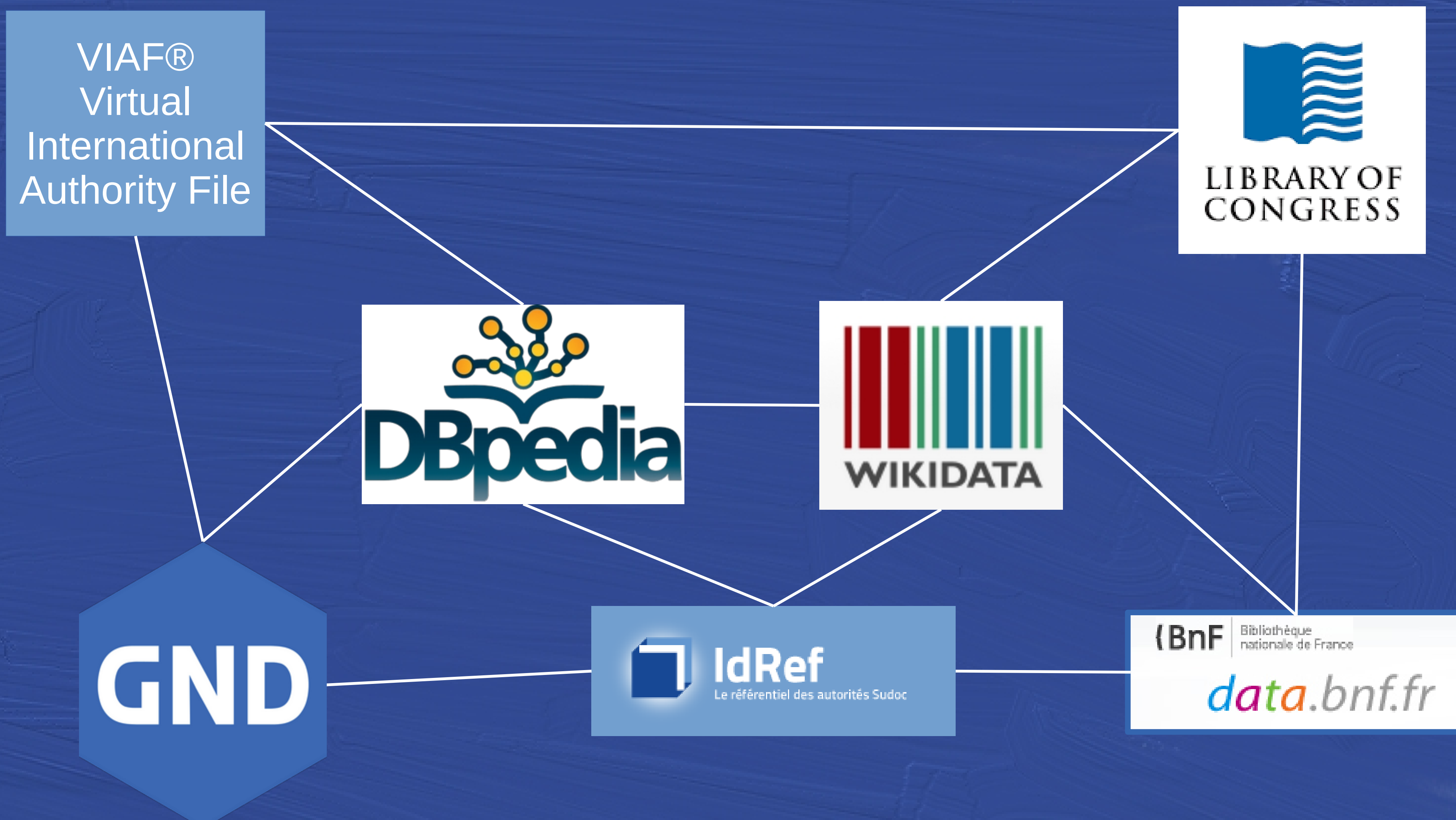
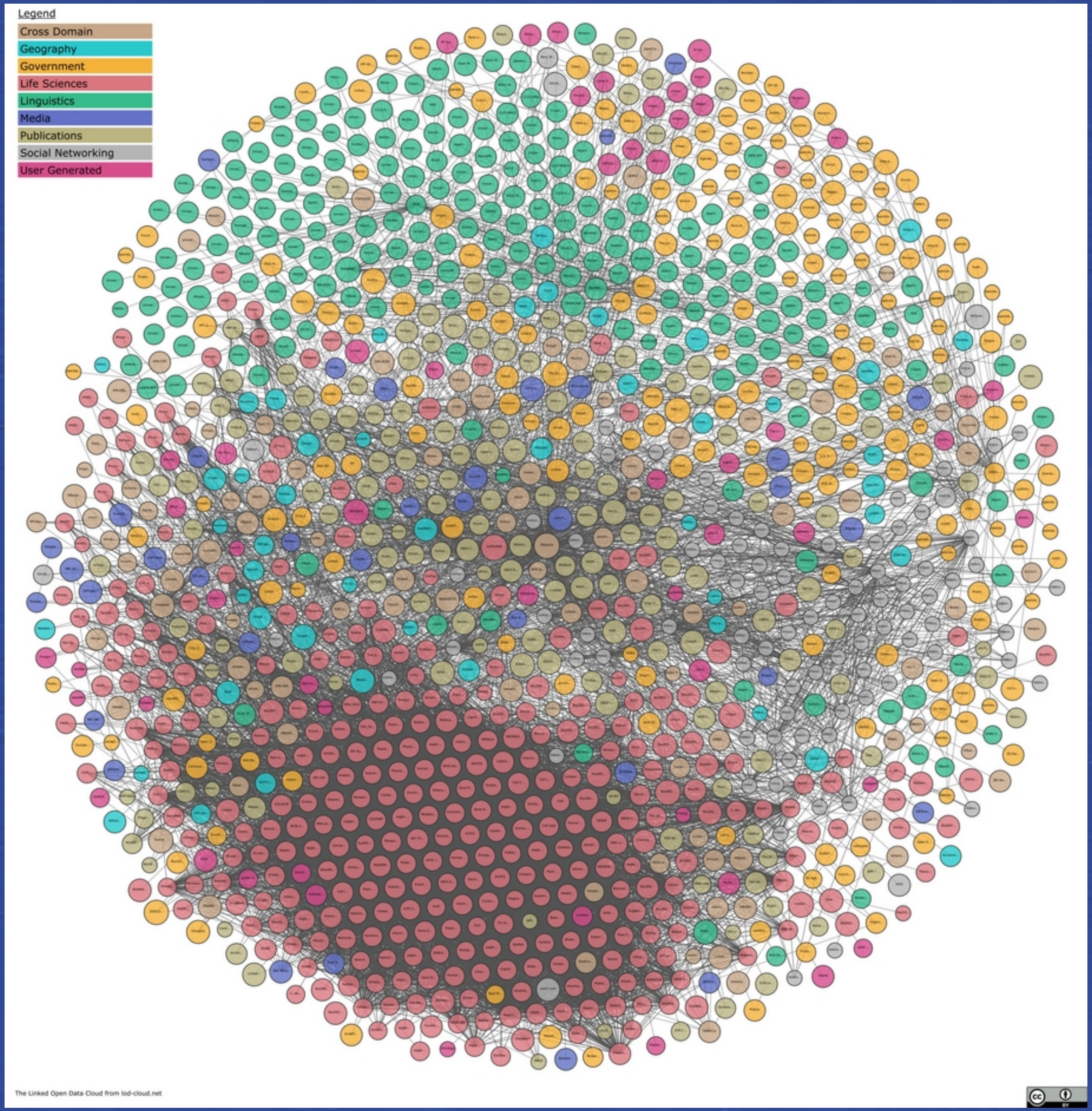
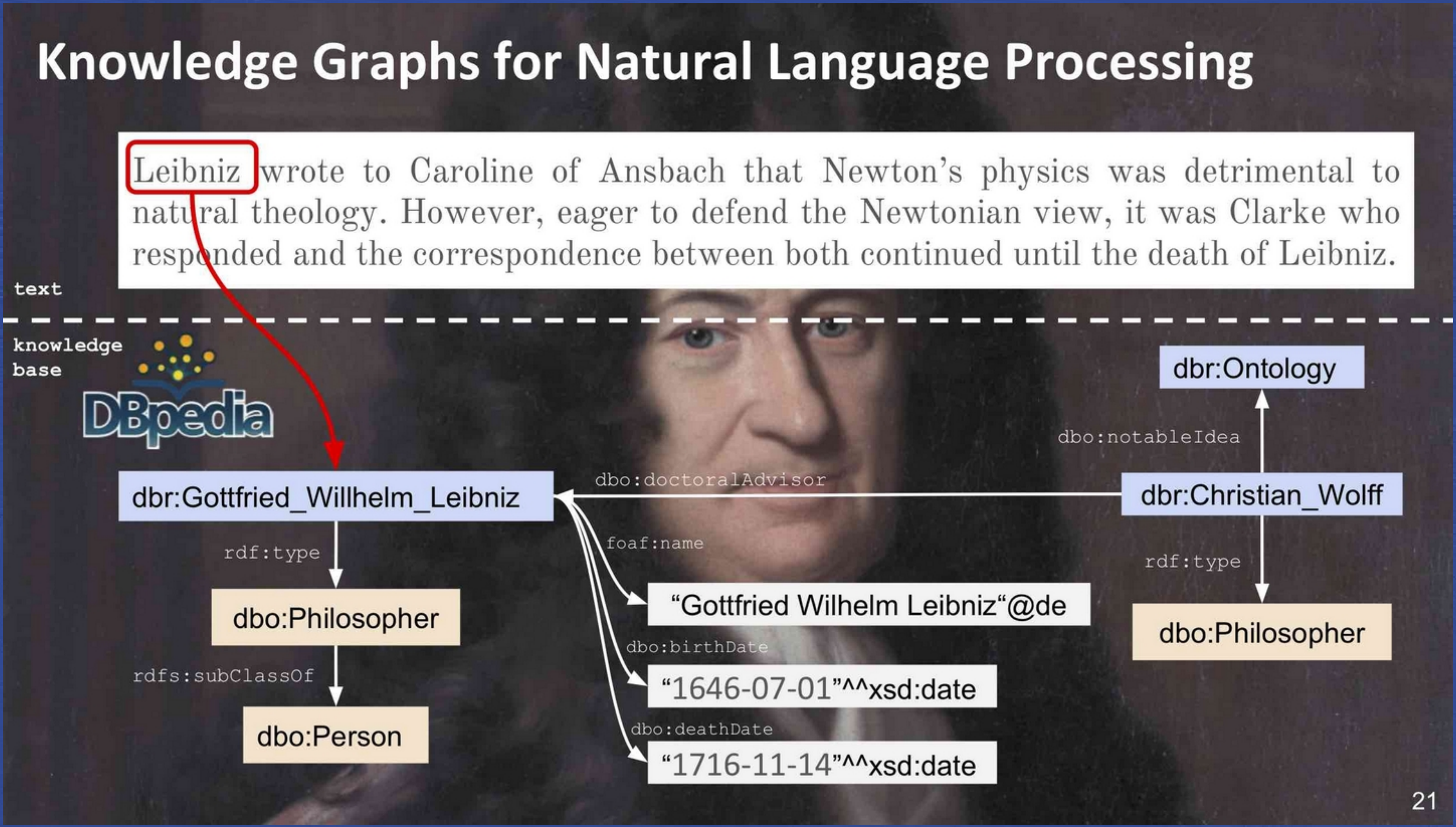


Semantic Web (Wikipedia)



https://lod-cloud.net/

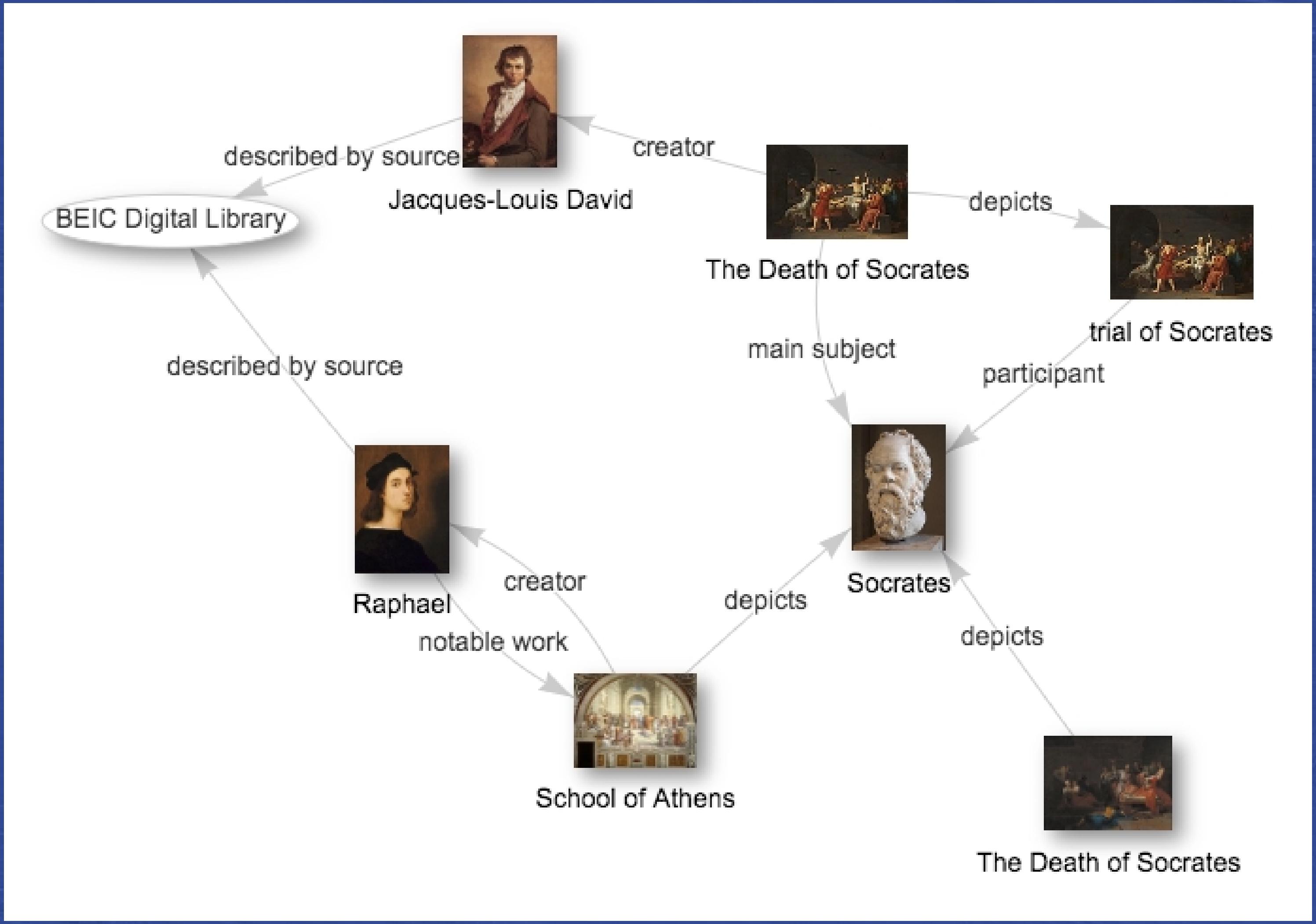


Karlsruher Institut für Technologie (29. Novembre 2017) – Antrittsvorlesung von

Prof. Dr. Harald Sack

Combining Semantics and Deep Learning for Intelligent Information Services

Wikidata : a semantic graph (*knowledge graph*) representing the objects in the world and their relationships

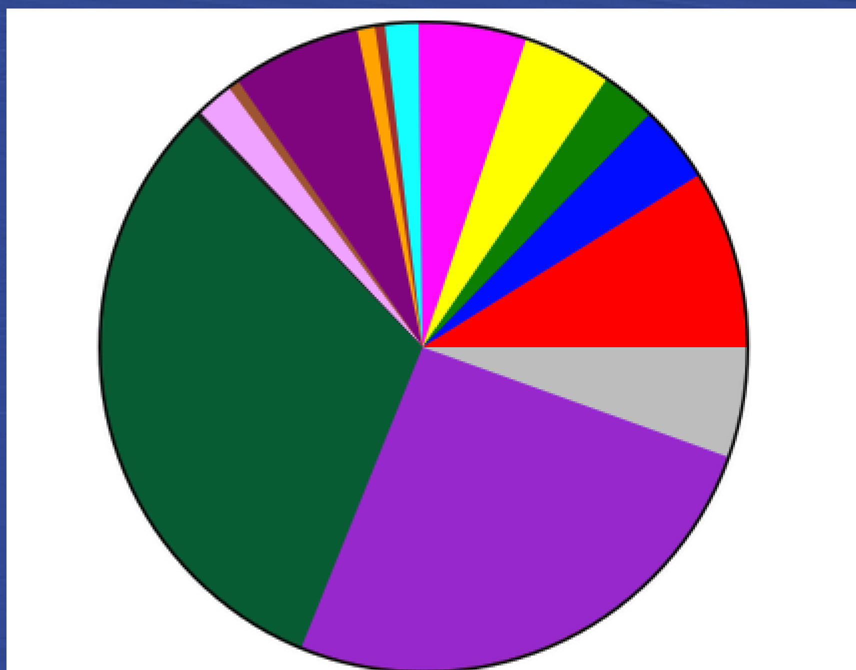


Source of the illustration : **click here - Try it out on Wikidata**

Wikidata

June 2026 – 122'170'394 items

8.8 billions statements

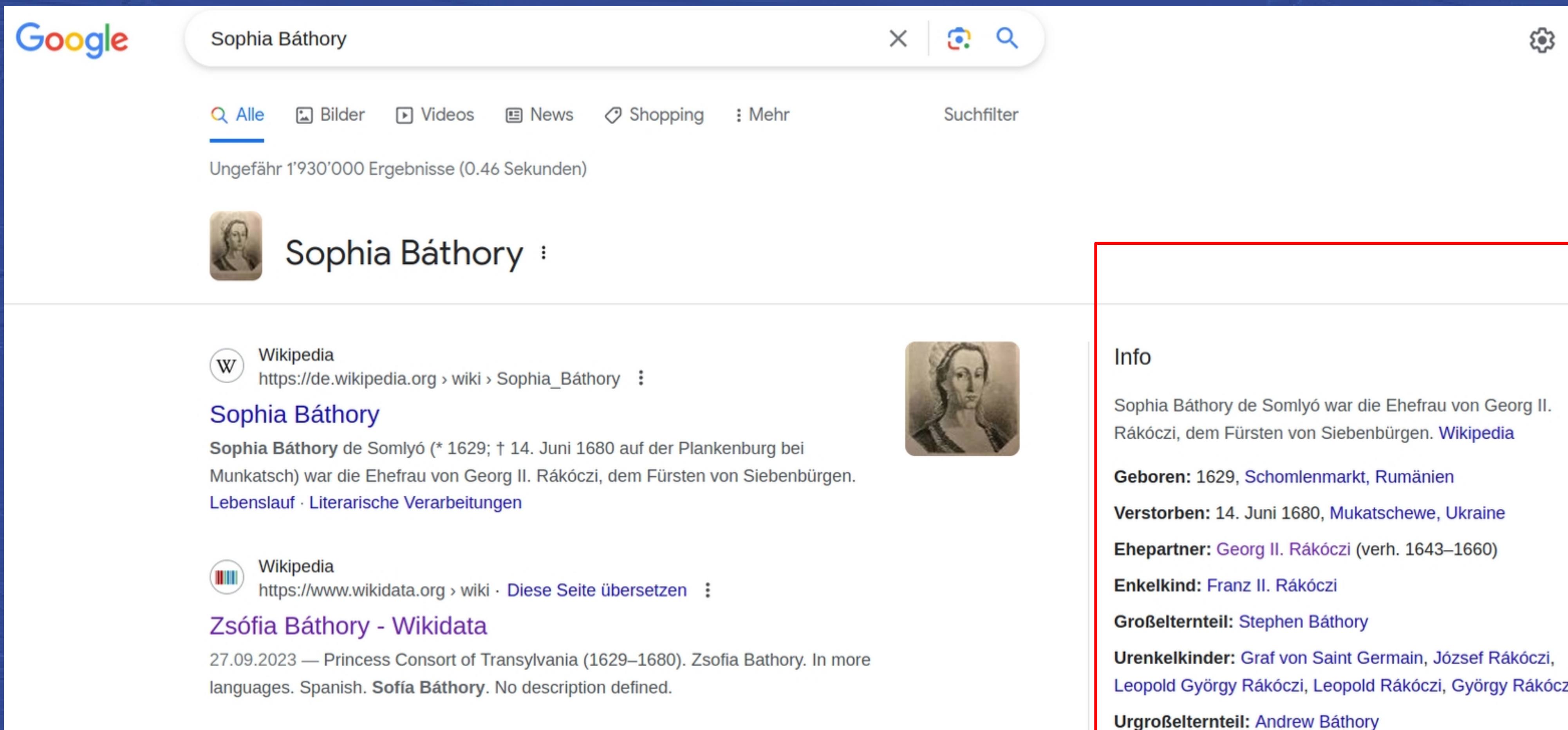


16 February 2020 : 71,611,020 items

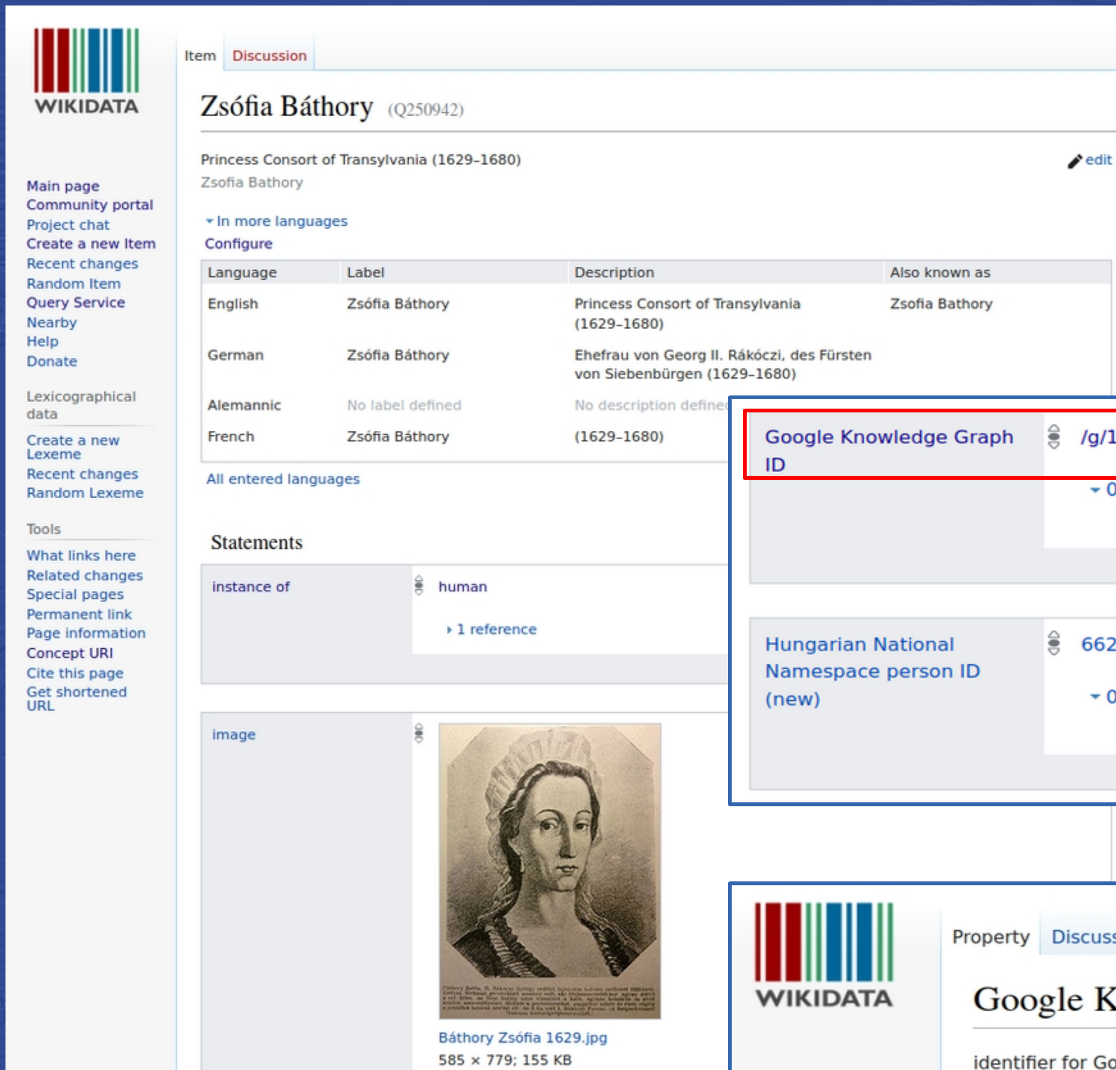
- human: 6,376,879 (8.9%)
- taxon: 2,726,046 (3.8%)
- administrative division: 1,943,285 (2.7%)
- architectural structure: 3,159,472 (4.4%)
- occurrence: 3,898,674 (5.4%)
- chemical compound: 1,188,724 (1.7%)
- film: 294,370 (0.4%)
- thoroughfare: 630,794 (0.9%)
- astronomical object: 4,601,733 (6.4%)
- Wikimedia list article: 404,454 (0.6%)
- Wikimedia disambiguation page: 1,358,230 (1.9%)
- Wikinews article: 195,900 (0.3%)
- scholarly article: 22,574,314 (31.5%)
- other P31/P279: 18,284,676 (25.5%)
- no P31/P279: 3,973,469 (5.5%)

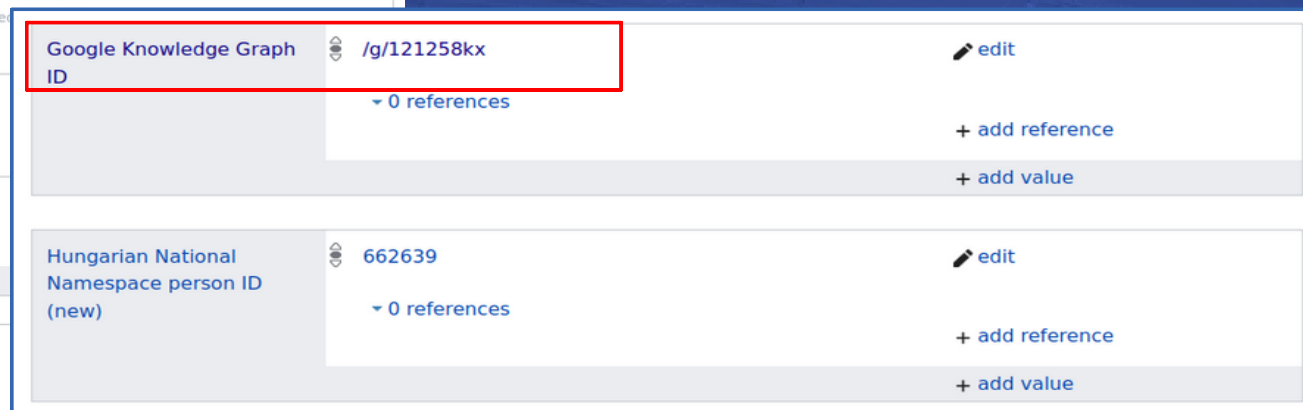
Google Knowledge Graph

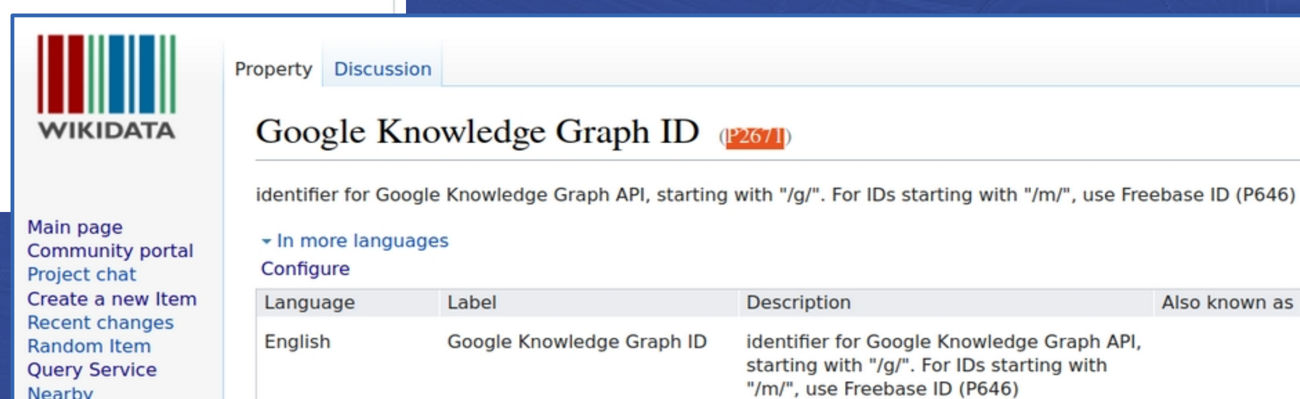
"By March 2023, it had grown to 800 billion facts on 8 billion entities" (Wikipedia).



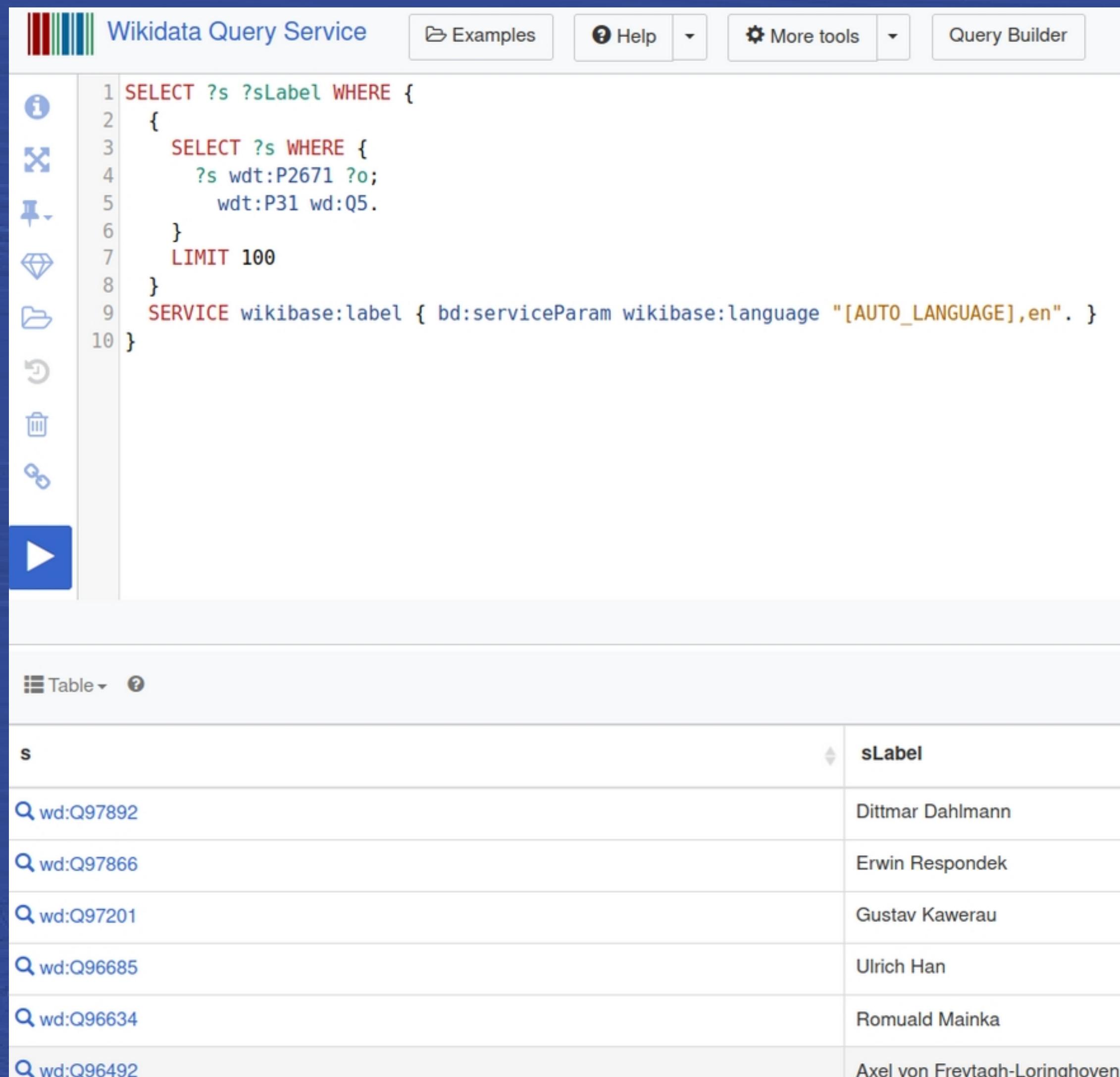
Wikidata and Google Knowledge Graph

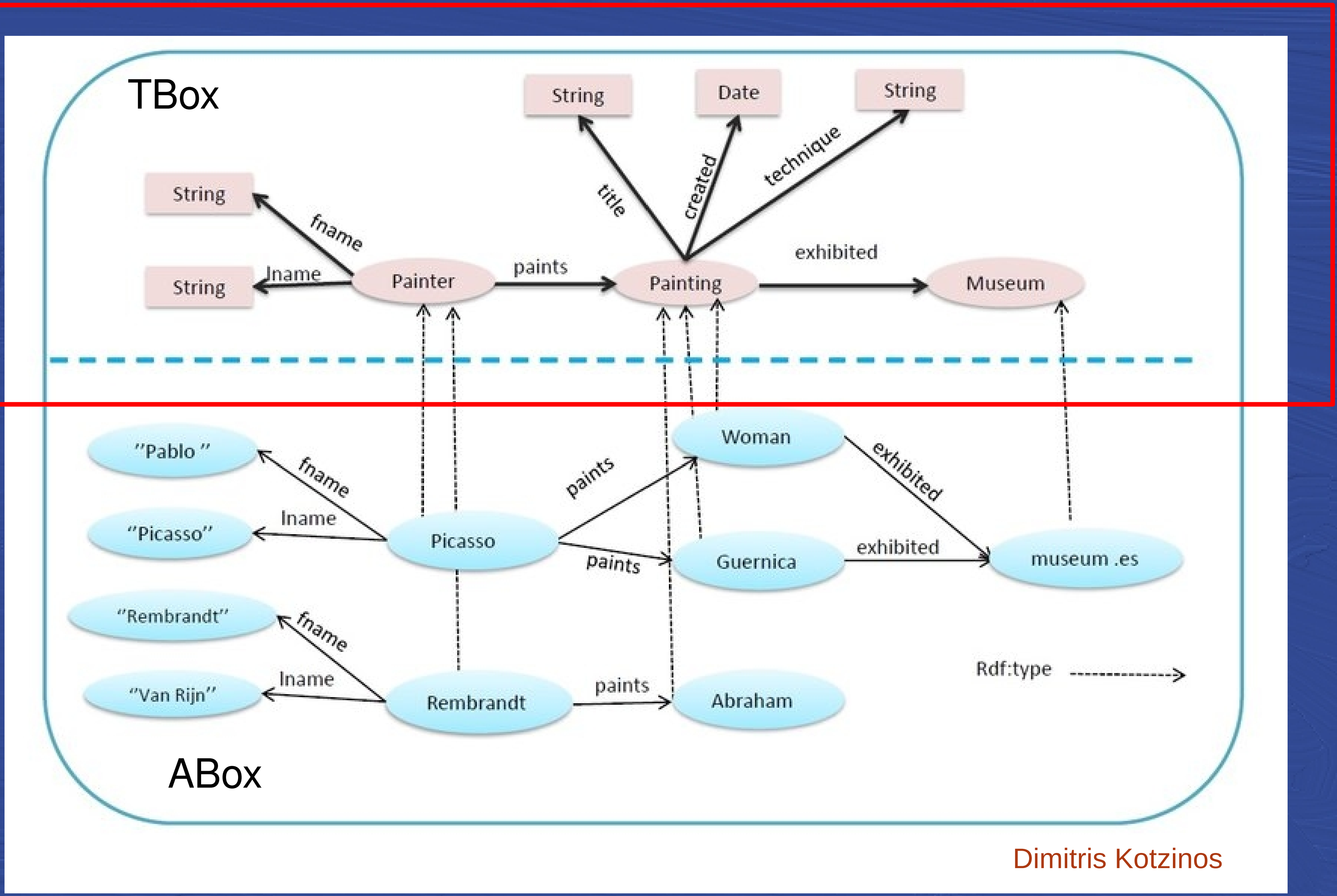






SPARQL : a query language for exploring LOD and the Semantic Web





Design a model of the research domain

Findable

Accessible

Interoperable

Re-usable

«There is an urgent need to improve the infrastructure supporting the ***reuse*** of scholarly data »

Wilkinson, Mark D., Michel Dumontier, Ijsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, et al. “*The FAIR Guiding Principles for Scientific Data Management and Stewardship*.” Scientific Data 3 (March 15, 2016): 160018.

The FAIR Data Principles

To be **Interoperable**:

I1. (meta)data use a *formal, accessible, shared, and broadly applicable **language** for **knowledge representation***.

I2. (meta)data use *vocabularies that follow FAIR principles*.

I3. (meta)data include qualified references to other (meta)data.

To be **Re-usable**:

R1. meta(data) have a plurality of accurate and relevant attributes.

R1.1. (meta)data are released with a *clear and accessible data usage license*.

R1.2. (meta)data are associated with their *provenance*.

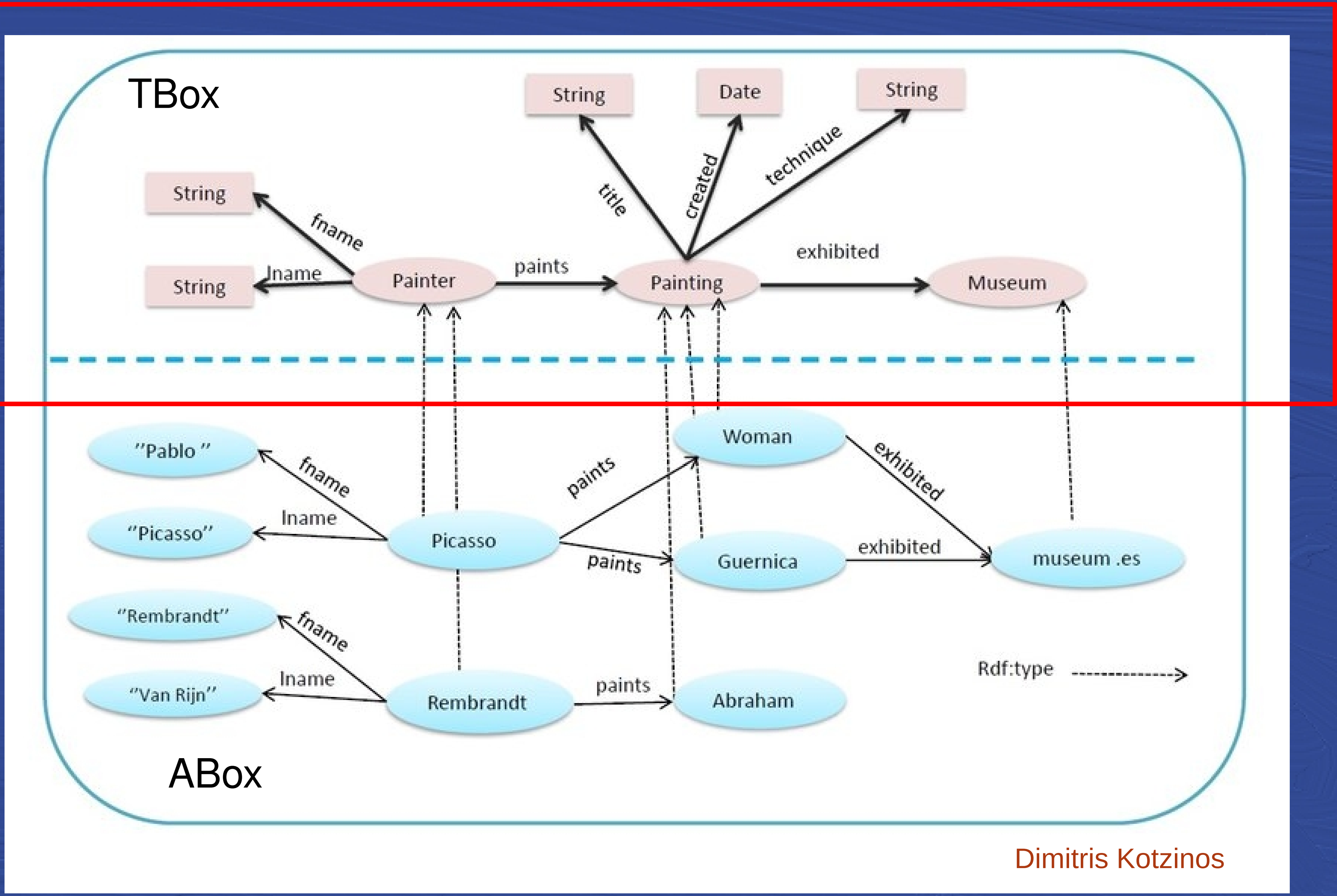
R1.3. (meta)data meet ***domain-relevant community standards***.

3.

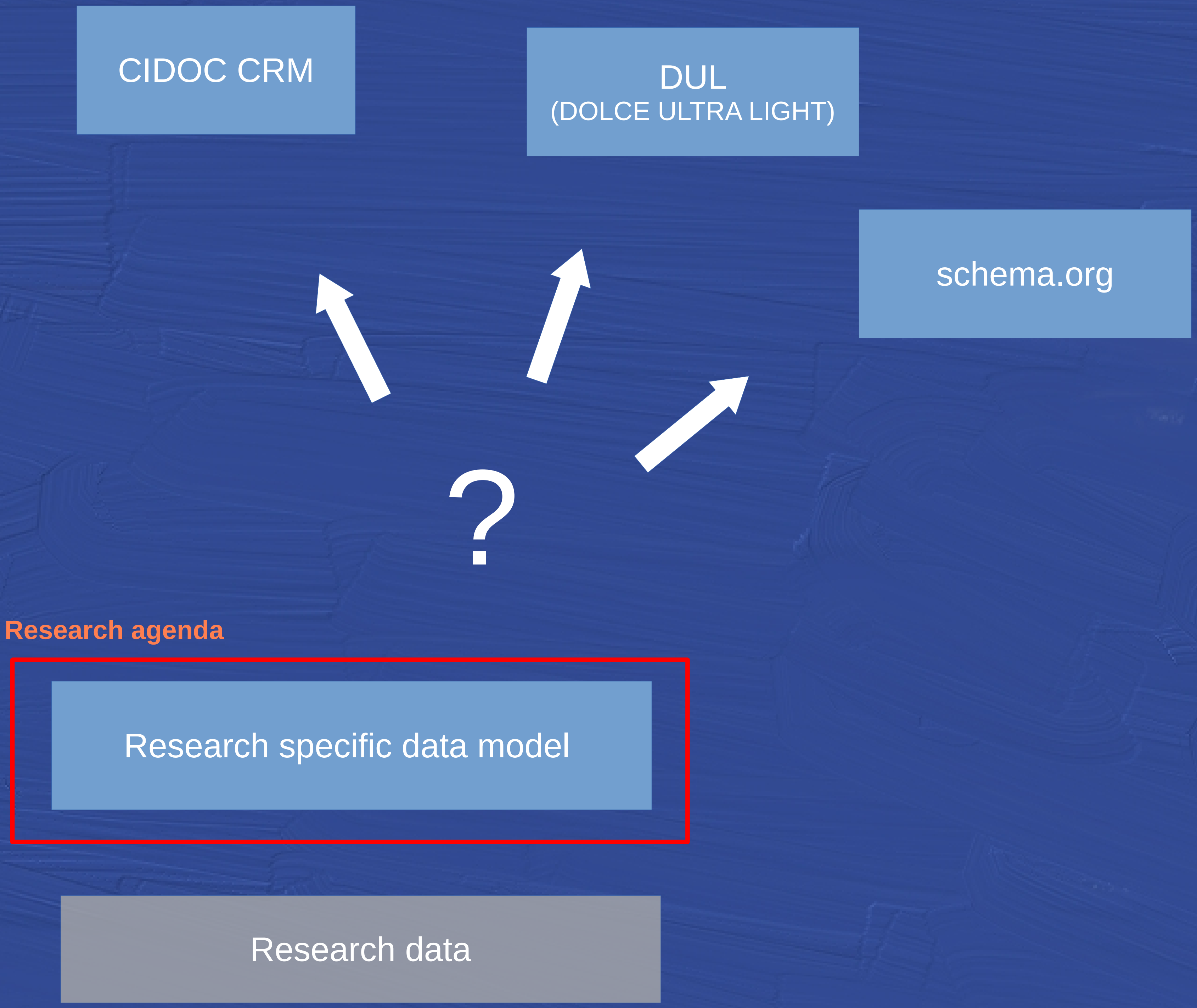
The endeavour :

developing a methodology,
creating infrastructure, building communities

The LOD4HSS initiative



Which model is suitable for HSS research ?


CIDOC CRM
DUL
(DOLCE ULTRA LIGHT)
schema.org
?
Research agenda
Research specific data model
Research data

Epistemological and methodological foundations

of the *Semantic Data for Humanities and Social Sciences* (SDHSS) ontology ecosystem

“An ontology is
a formal explicit specification
of a shared conceptualization
of a domain of interest”

- « Formality – ... a knowledge representation language that is based on the grounds of **formal semantics**. »
- « Consensus – … an agreement on a domain conceptualization among people in a community. »
- « Conceptuality – … in terms of conceptual symbols that can be intuitively grasped by humans, as they correspond to the elements in their **mental models**. »
- « Domain Specificity – ... limited to knowledge about a particular **domain of interest**. »

[Domingue et al. 2011, p. 510-511]

What are scientific disciplines?

- Communities of knowledge that share a **common domain of discourse** and **of scientific interest**.

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What are scientific disciplines?

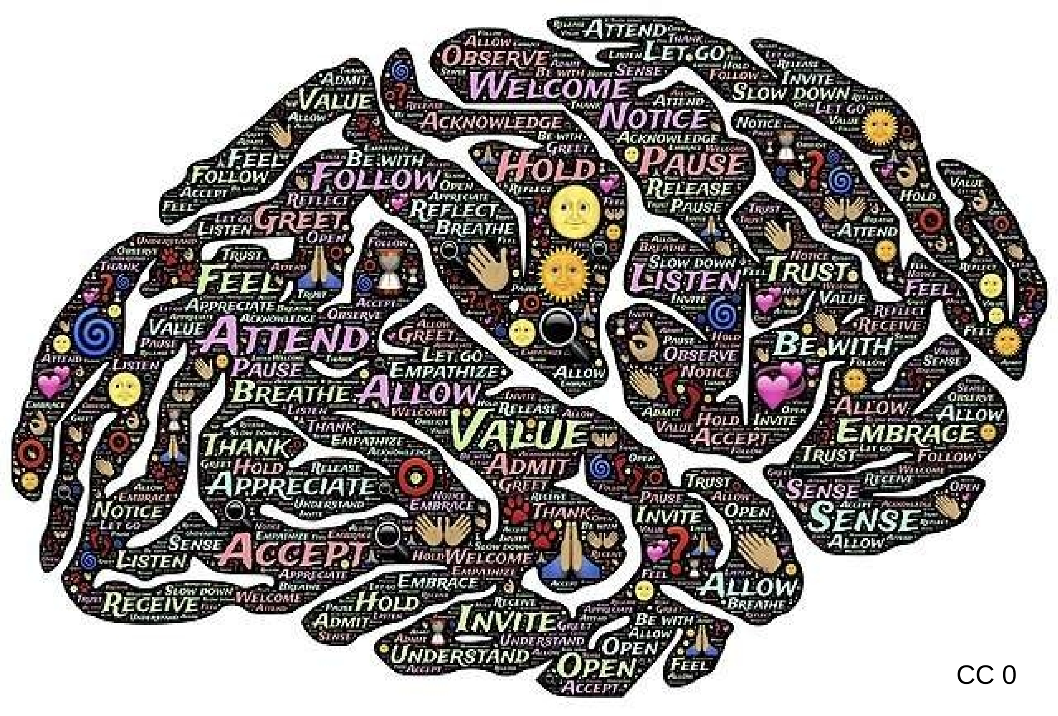
- Communities of knowledge that share a **common domain of discourse** and **of scientific interest**.
- They define scientific objects and research questions that are considered as valid in a scientific domain.
- They share a specific methodology which must be empirical and provide reproducible results.
- They produce new **knowledge** in their **domain** and this knowledge may have more or less impact on human societies in which they are embedded.

The domain of HSS

Material and biological world

Mental reality

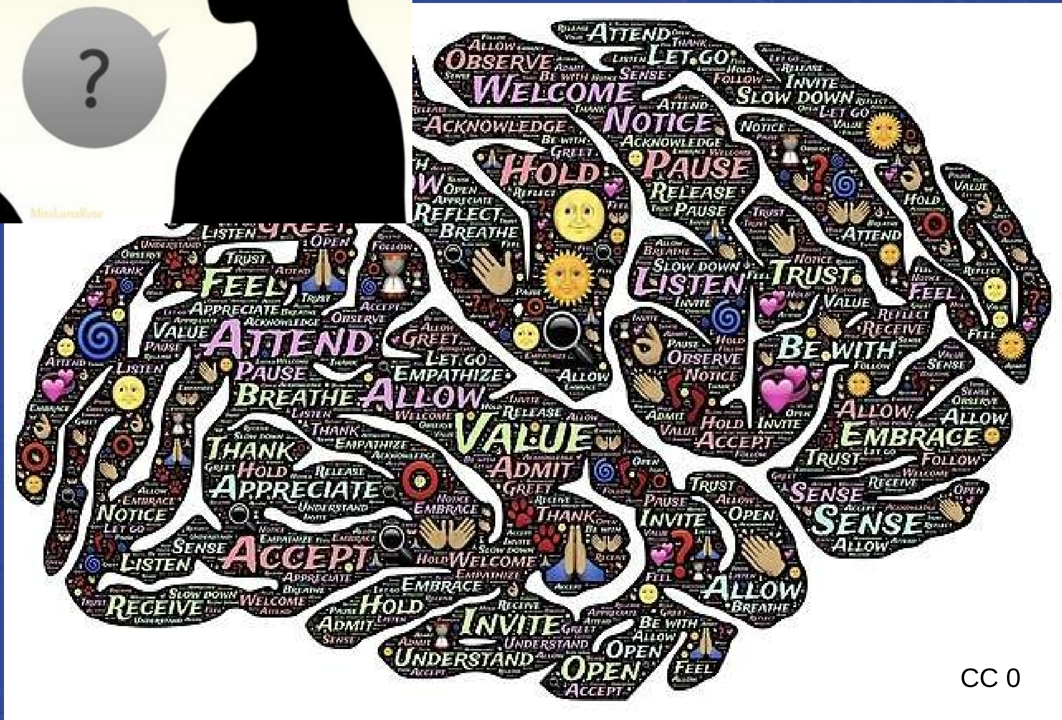
Material
and biological
world



Social reality

Mental reality

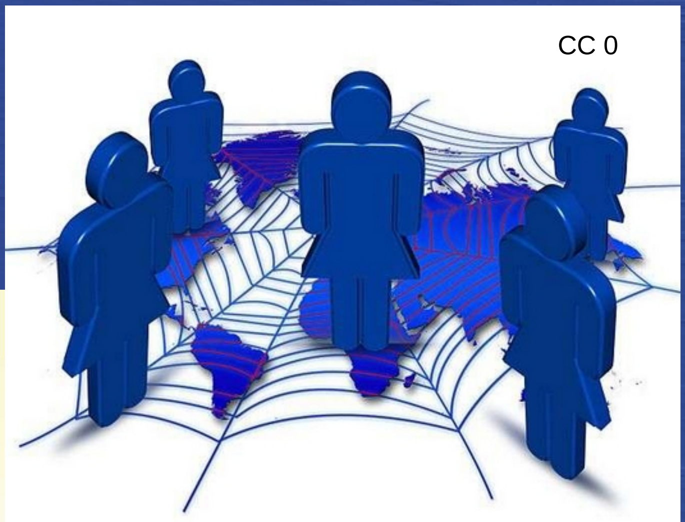
Material and biological world

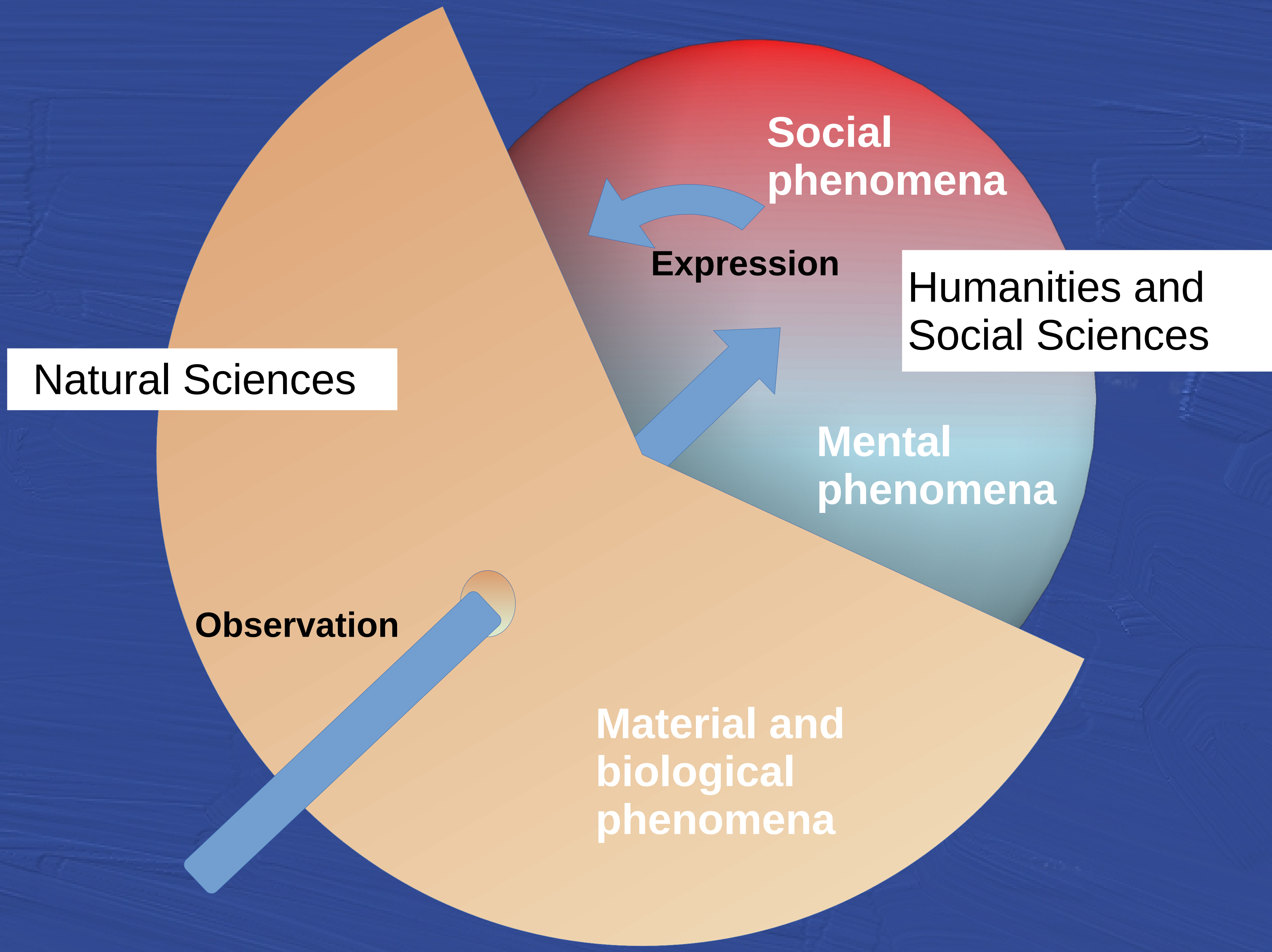


Social reality

Mental reality

Material
and biological
world




Social phenomena
Expression
Humanities and Social Sciences
Natural Sciences
Mental phenomena
Observation
Material and biological phenomena

How can we conceptualized observable mental and social phenomena in HSS disciplines ?

How are conceptualized observable mental and social phenomena in HSS disciplines ?

- Social representations (social sciences):

  «social identity emerges based on social representations ... Identity is a way of organizing meanings, of being constructed as a social subject»

  Álvarez Bermúdez, Javier and Juana Juárez-Romero. "Identities, Memory and the Construction of Citizenship." Papers on Social Representations 28, no. 2 (2019).

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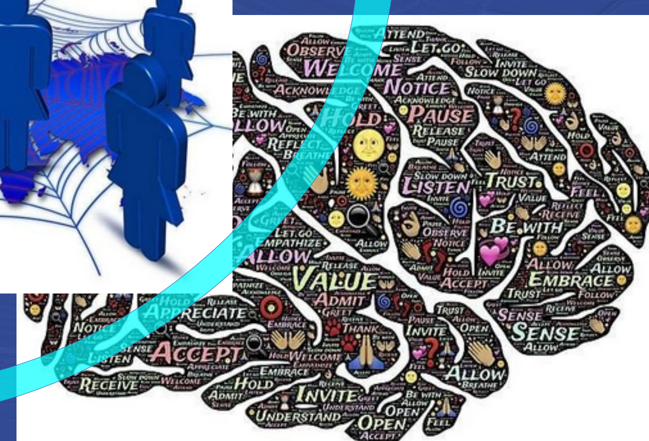
- Collective intentionality (social philosophy):

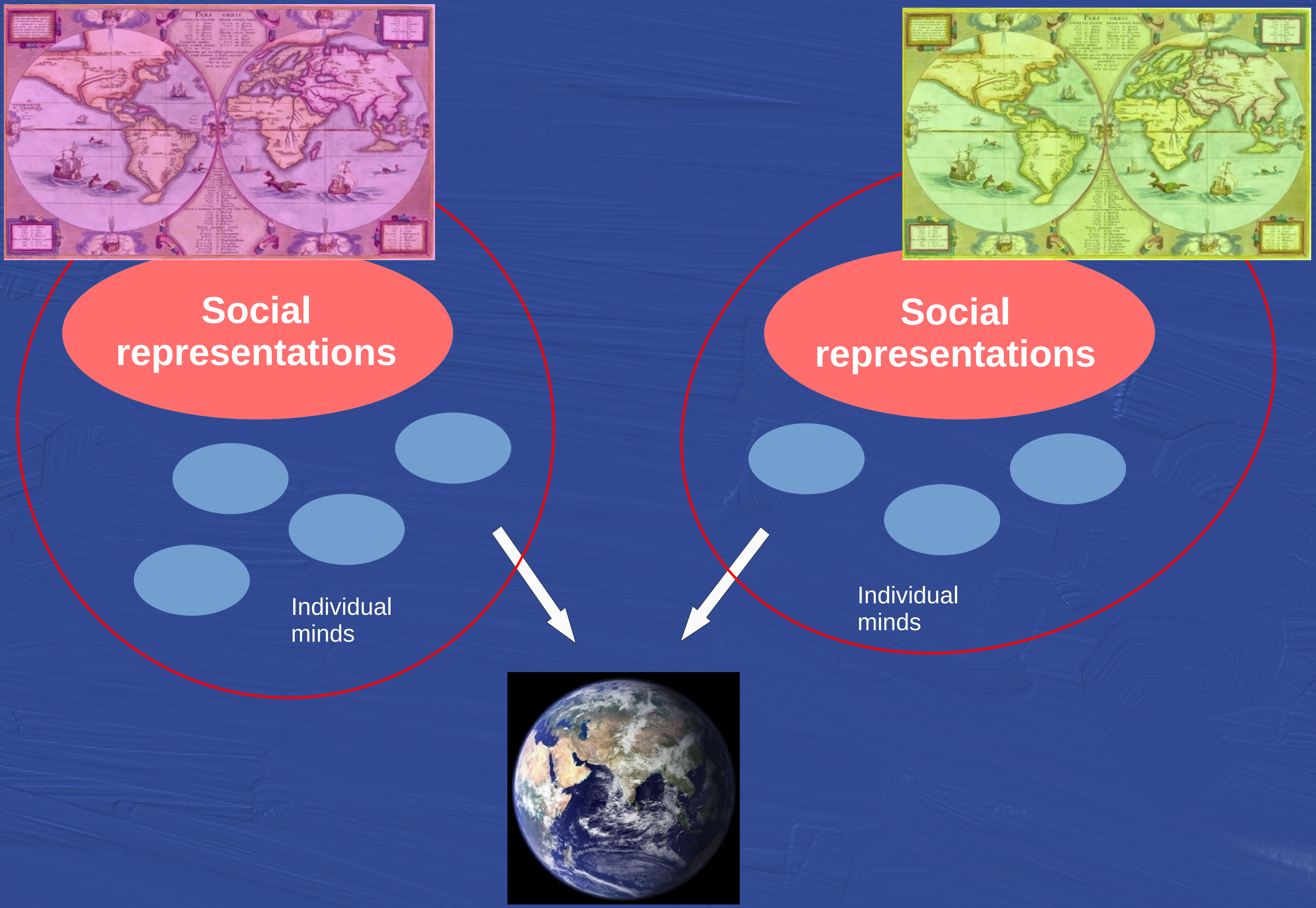
  «Consciousness and intentionality are caused by and realized in neurobiology. Collective intentionality is a type of intentionality, and society is created by collective intentionality. ... language enables the creation and continue dexistence of status functions that do not require any physical existence beyond thelinguistic representations themselves. »
  John Searle. Making the Social World: The Structure of Human Civilization, Oxford, 2010.


PARS ORBIS

CC 0

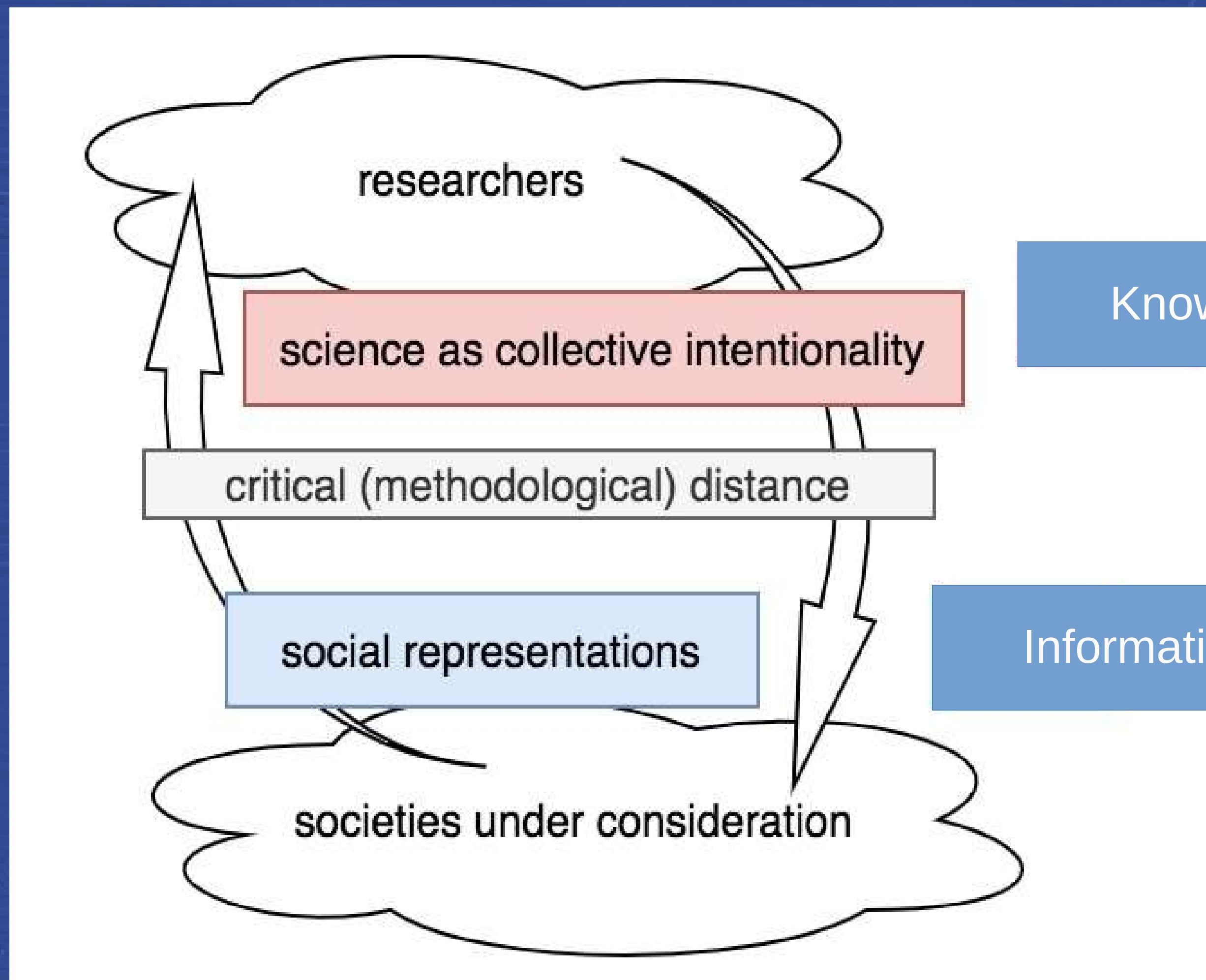

WELCOME
NOTICE
PAUSE
LISTEN
TRUST
BE WITH
VALUE
ACCEPT
EMBRACE
SENSE
ALLOW
INVITE
RECEIVE
UNDERSTAND
OPEN

CC 0


Social representations
Individual minds
Social representations
Individual minds

Information as representations of the world

Knowledge as interpretation of the world



Knowledge

Information

Information as **representation** of the world

Social Representations (Collective Intentionality)

An **ontology** as a **representation** of the world :

- representation of the **things (→ objects)** in the world

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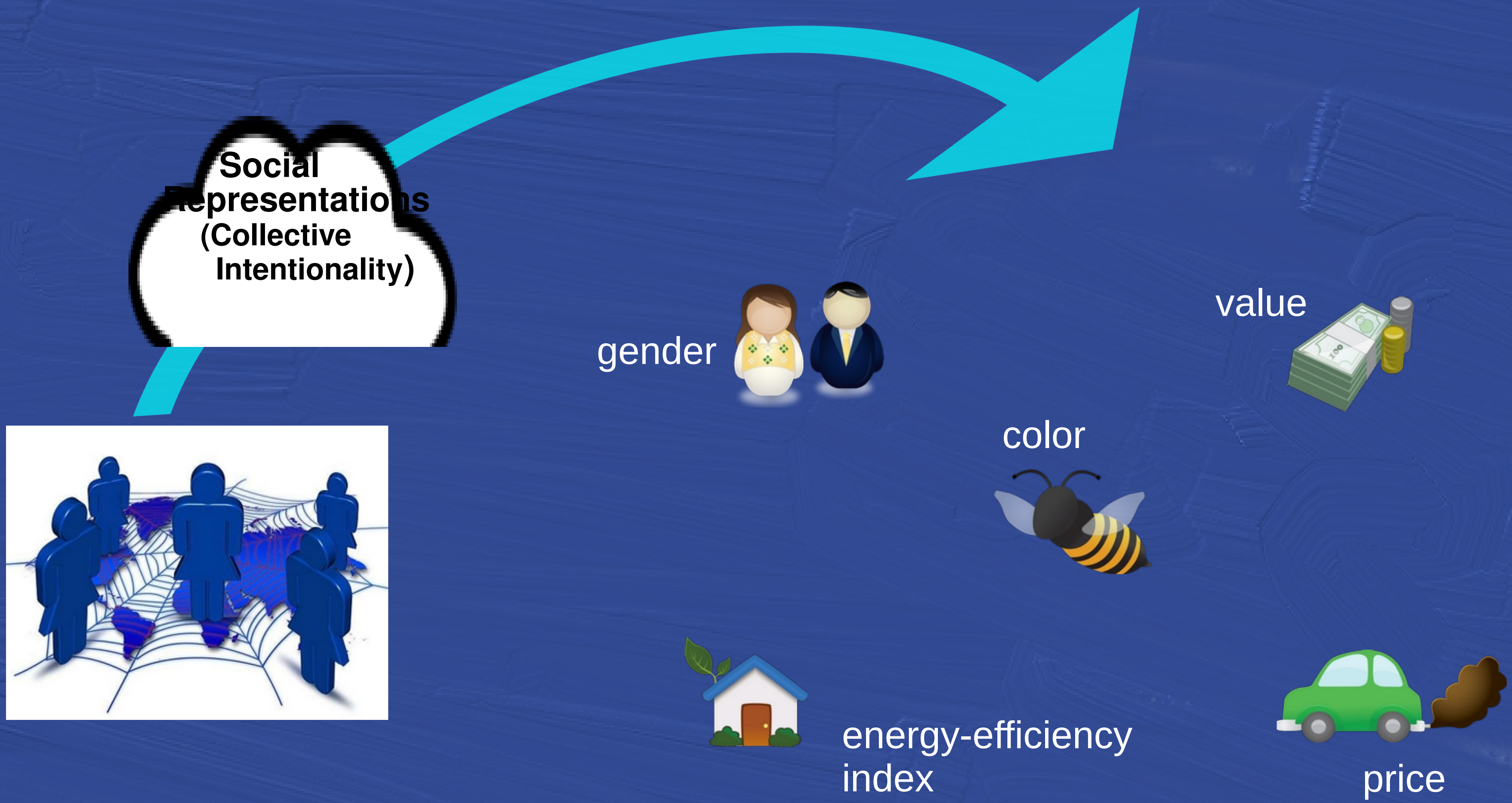
- of their **properties** (qualities)



An **ontology** as a **representation** of the world :

- representation of the **things (→ objects)** in the world
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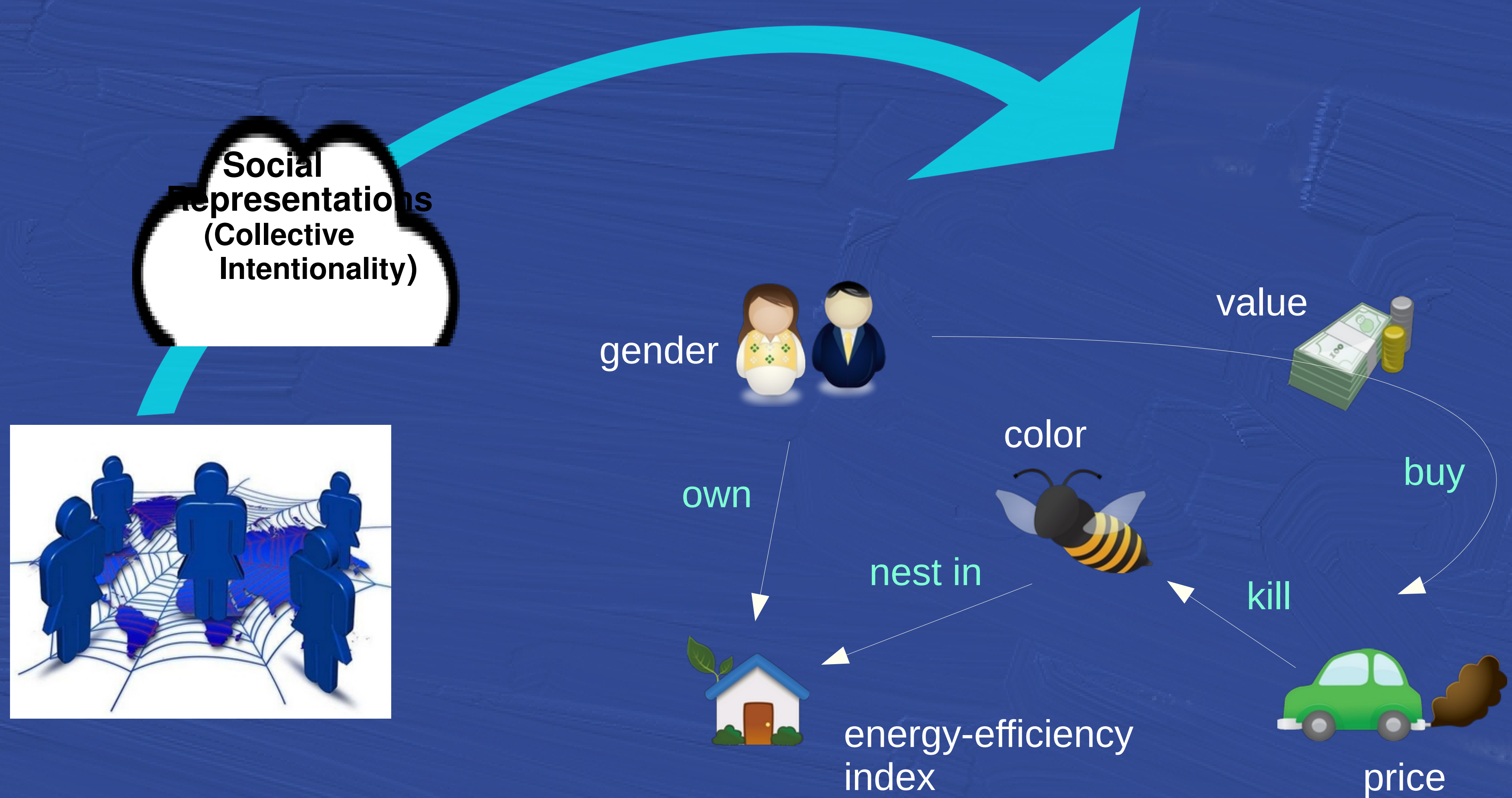
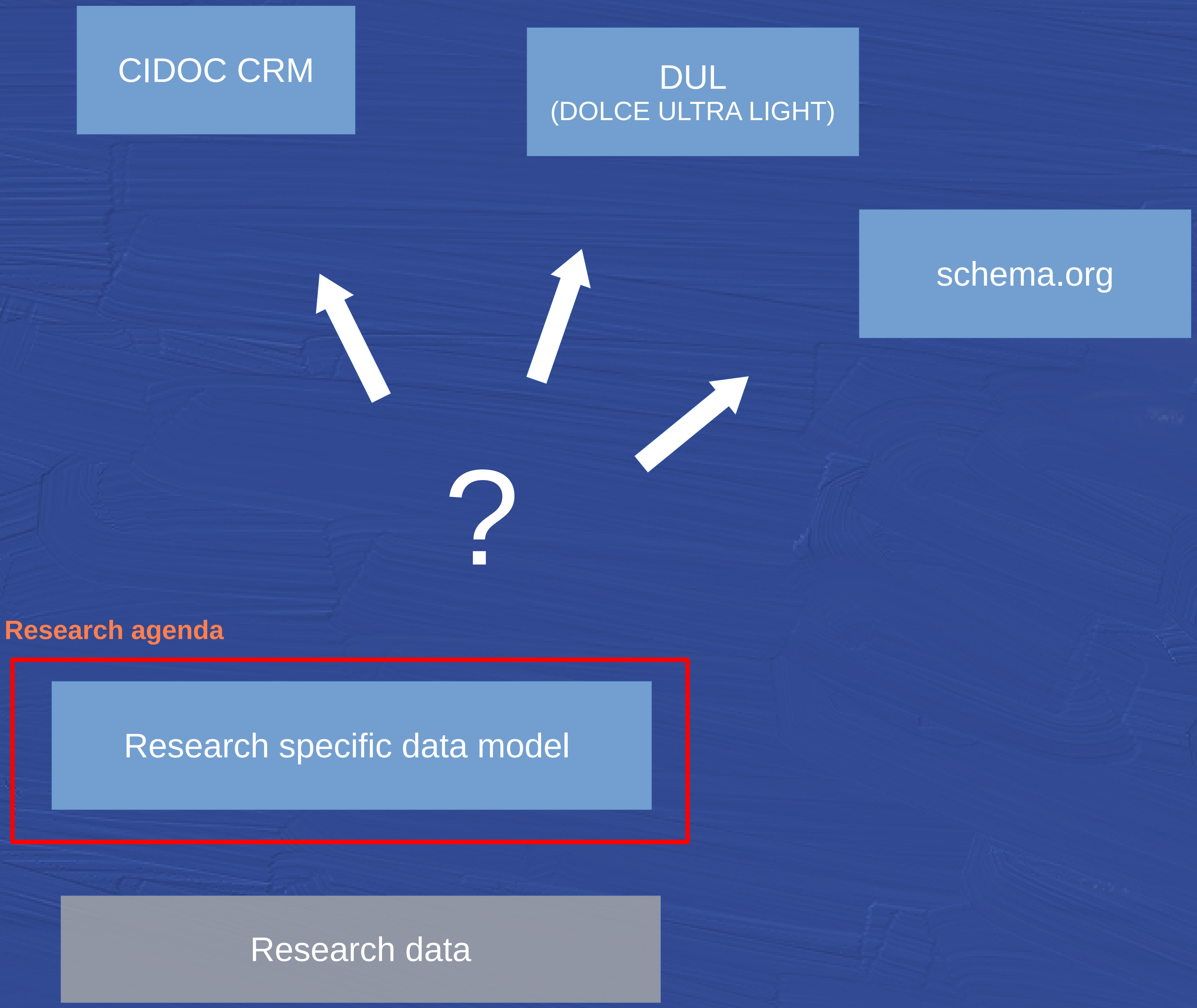
- of their **relationships**




CIDOC CRM
DUL
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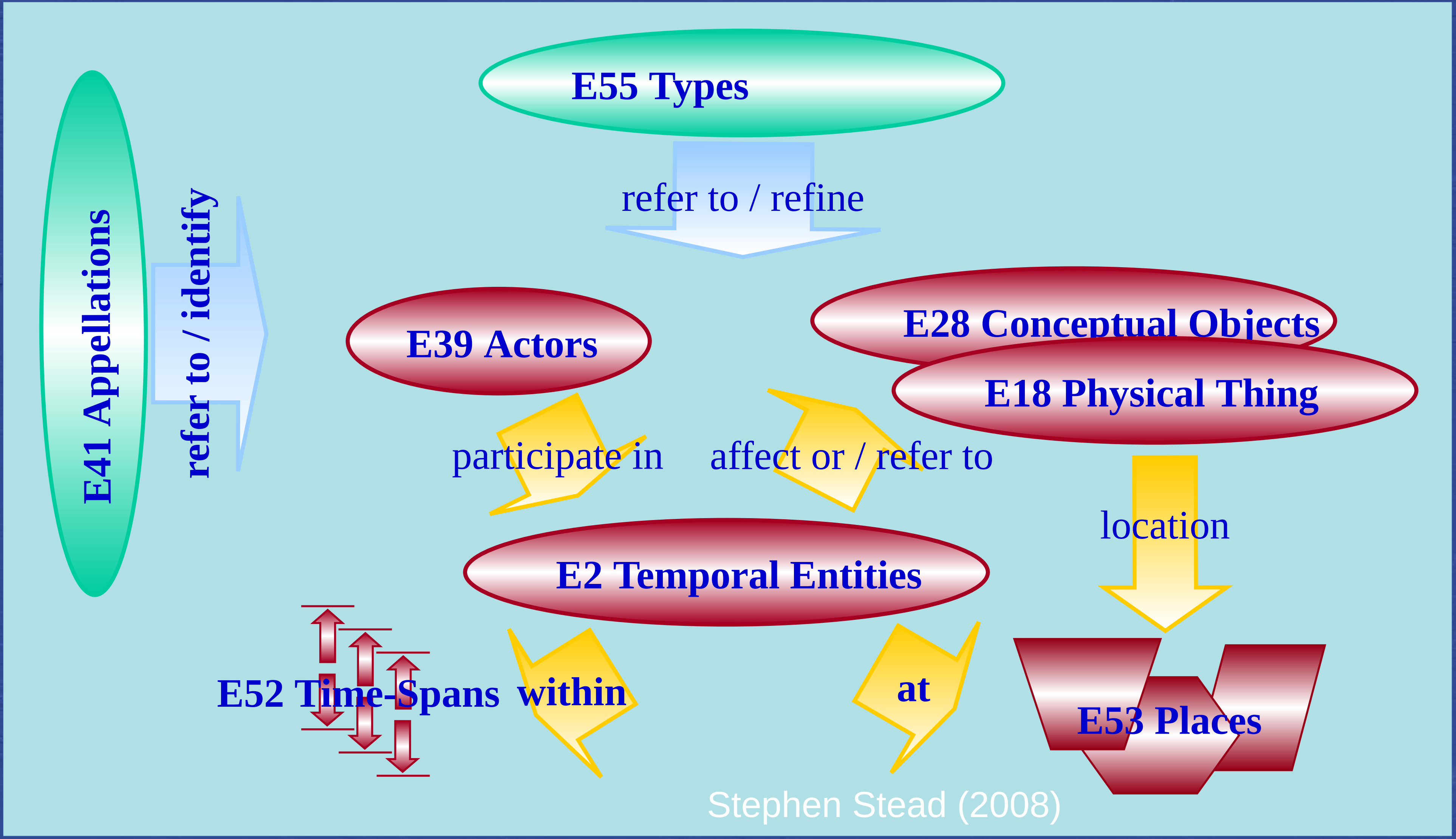
schema.org
?
Research agenda
Research specific data model
Research data

The CIDOC CRM (ISO21127:2023 – community edition 7.1.3)

A semantic framework that conceptualises spatio-temporal phenomena

and provides *interoperability* in the domain of cultural heritage information



Foundational ontologies
& modelling best practices

DOLCE + Descriptions and Situations
& object-oriented modelling principles

Research agenda

Research specific data model

Research data

Ontology engineering as a relevant

research domain in computer science

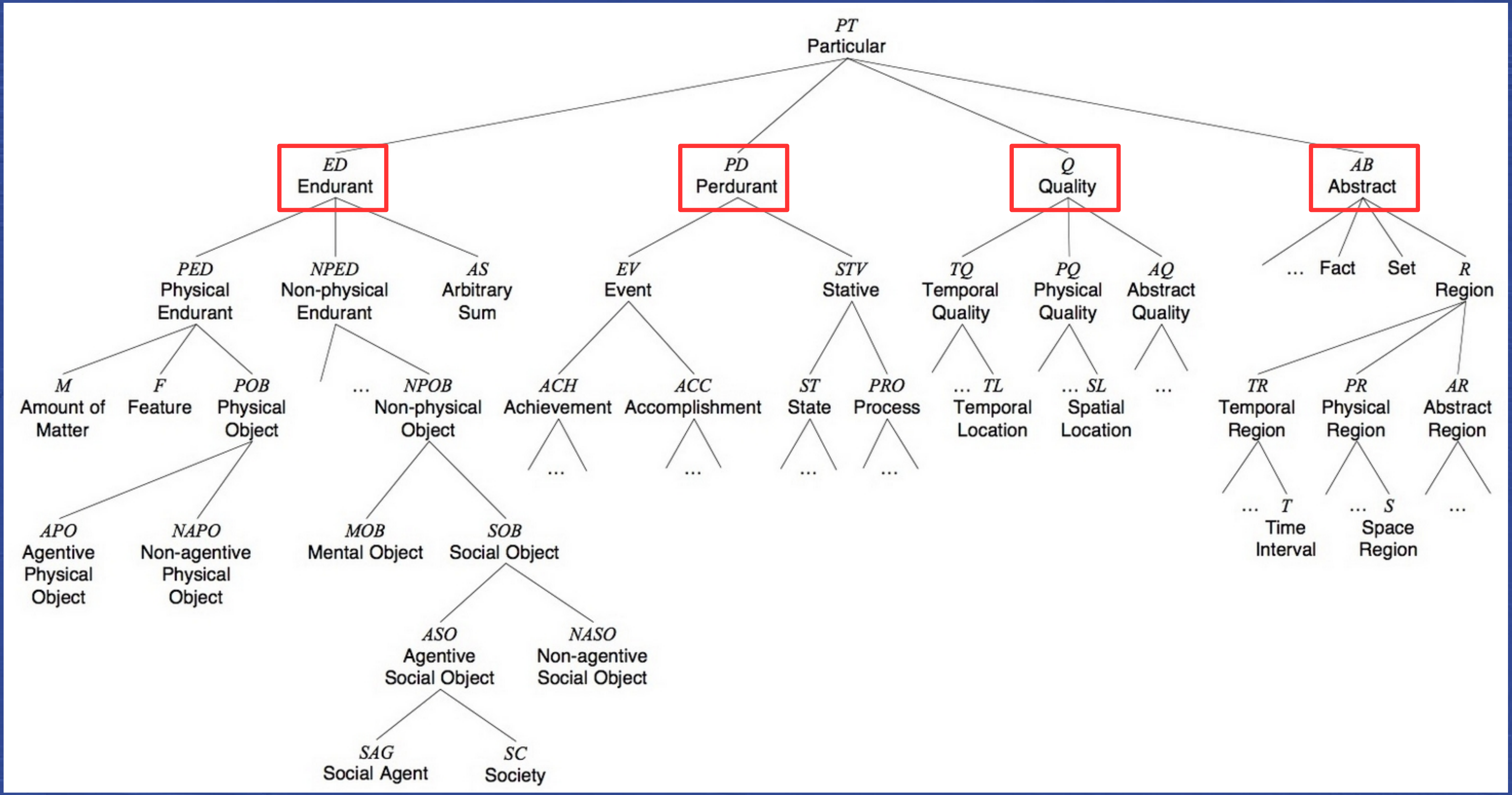
notably regarding top-level or **foundational** ontologies

« **Foundational ontologies** are not directly used for applications; rather, they provide **conceptual handles** to solve cases of misunderstandings due to the limitations of expressiveness of the application languages. […]

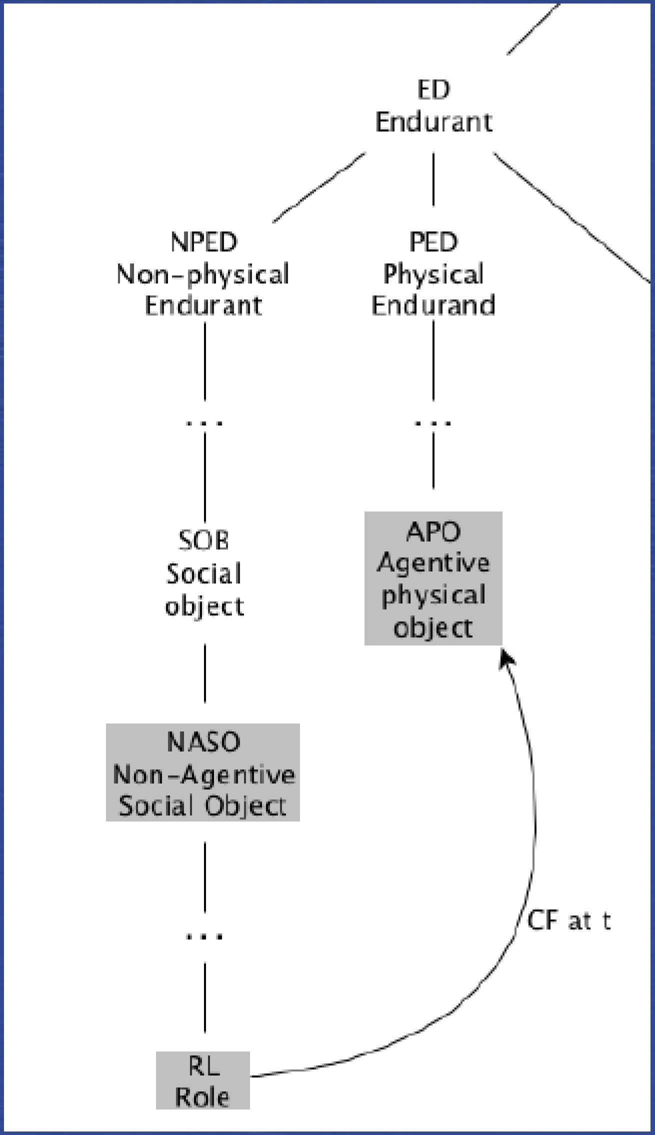
DOLCE has remained fixed over the years fulfilling the purpose of top-level ontologies to provide a solid and stable basis for **modeling different domains**, in this way **ensuring interoperability of reference and domain ontologies** that use DOLCE. »

Borgo Stefano et al., « DOLCE: A descriptive ontology for linguistic and cognitive engineering1 », Applied Ontology, 18.11.2021, pp. 1-25.

Foundational ontologies
were developed to support the verification and improvement of the **conceptualization** of a **domain of discourse.**



Descriptive Ontology for Linguistic and Cognitive Engineering (DOLCE) – a foundational ontology designed in 2002 in the context of the WonderWeb EU project, developed by Nicola Guarino and his associates at the Laboratory for Applied Ontology (LOA) – WonderWeb Deliverable D18, p.14



∀ x ¬CF(x, 2 CT eacher, t2) ∧
CF(Potter, 2 CTeacher, t1) ∧
CF(Bumblebee, 2CTeacher, t3)

CF(Mary, 2CStudent, t1)

Borgo Stefano et al., « DOLCE: A descriptive ontology for linguistic and cognitive engineering », Applied Ontology, 18.11.2021, pp. 1-25.

Foundational ontologies
& modelling best practices

DOLCE + Descriptions and Situations
& object-oriented modelling principles

Generic, domain related core ontology

CIDOC CRM

Research agenda

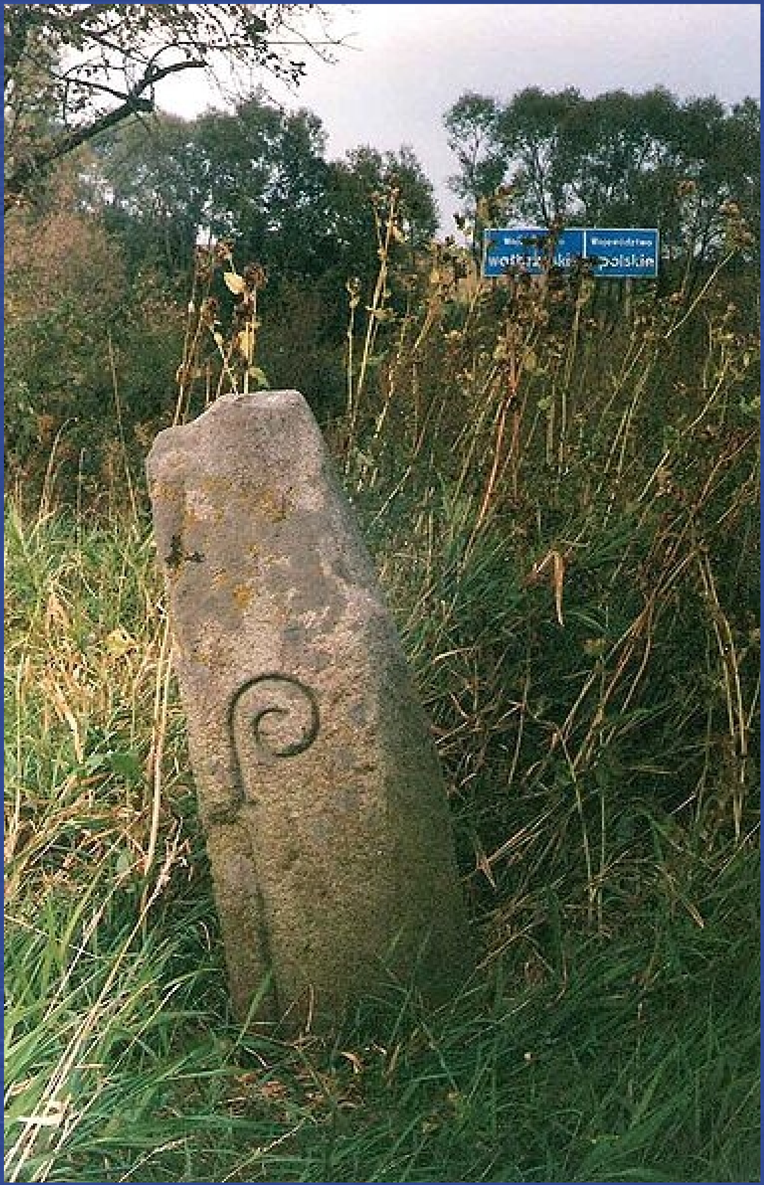
Research specific data model

Research data

Examples of objects and questions

in the HSS domain

Artefacts and their social significance: a border stone represents a delineation of a border



Border stone from c. 1280 on the border of the lands of the Wrocław bishops.

In the background the contemporary road sign marking the border between Opole and Wałbrzych voivodships.

Photo: Władysław Łoś (CC By SA 3.0)

The physical object, the notion of border, the symbol of the episcopal crozier, the function of bishop.

The bio-physical world, the social world

Social roles: the rite that establishes the function

Different perspectives on the same events: interpreting a car race as fun or as a crime

Sports car photo created by azerbaijan_stockers - www.freepik.com

Foundational ontologies
& modelling best practices

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Generic, domain related core ontology

CIDOC CRM

SDHSS

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Semantic Data for Humanities and Social Sciences (SDHSS) CIDOC CRM Core Extension

Semantic Data for Humanities and Social Sciences (SDHSS) CIDOC CRM Top-Level Extension

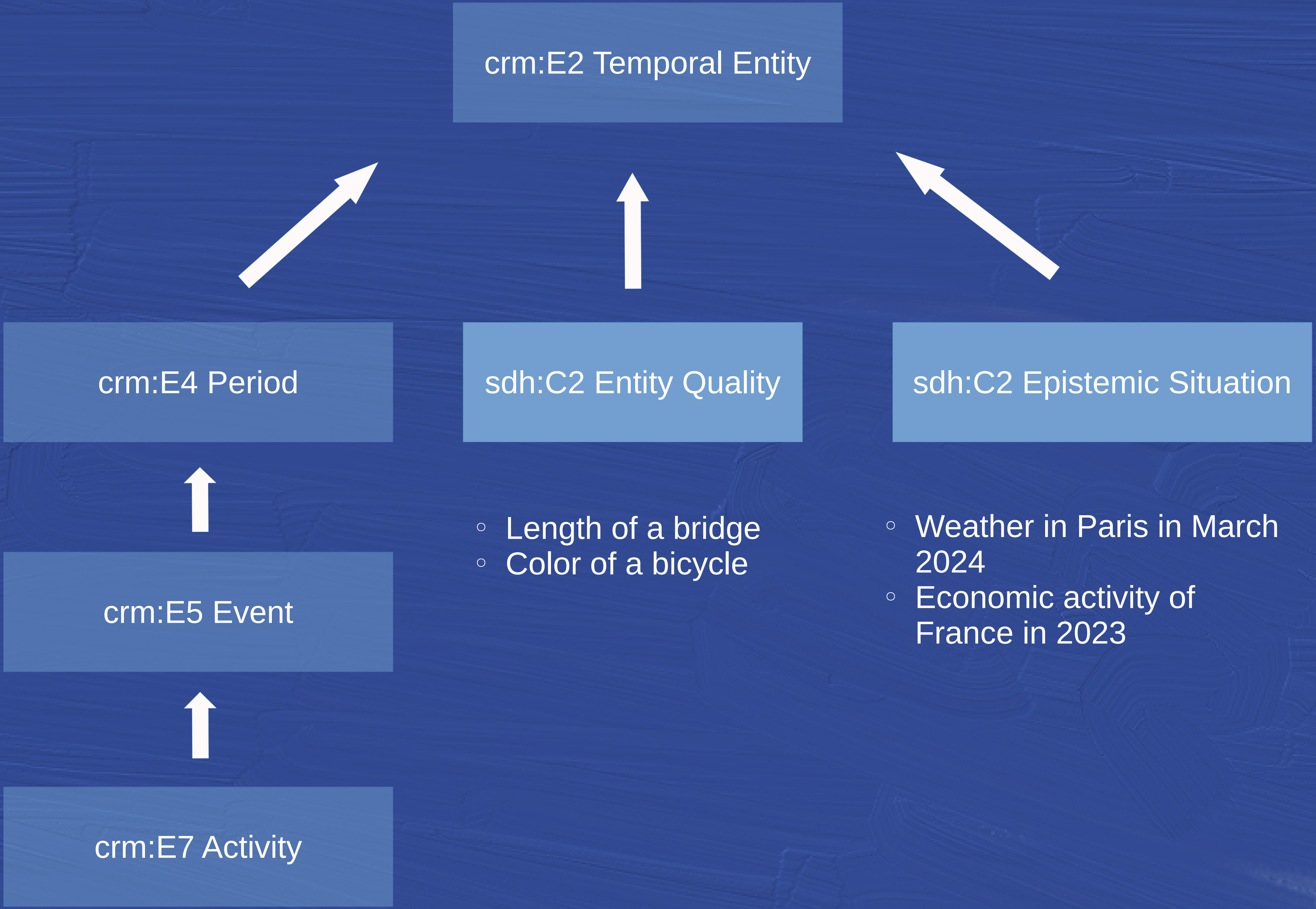
Description:

Published by Francesco Beretta (CNRS/Université de Lyon), 7 December 2020. Last revised on March 30 2021. (CC BY-SA 4.0)

The extension of CIDOC CRM for semantic data for humanities and social sciences (SDHSS) stems from the need to conceptualise the reality in the world, and more specifically factual information, from the point of view of historical research. The ontological commitment is therefore related to the domain of discourse of history but insofar as history, as a discipline that studies the life of humans and societies in the past, is interested in all the different aspects of social, economic, political, religious, literary and cultural life, the scope of this extension could be defined as the whole of social and human life, apprehended from the descriptive point of view, and global approach to reality, that characterises historical research.

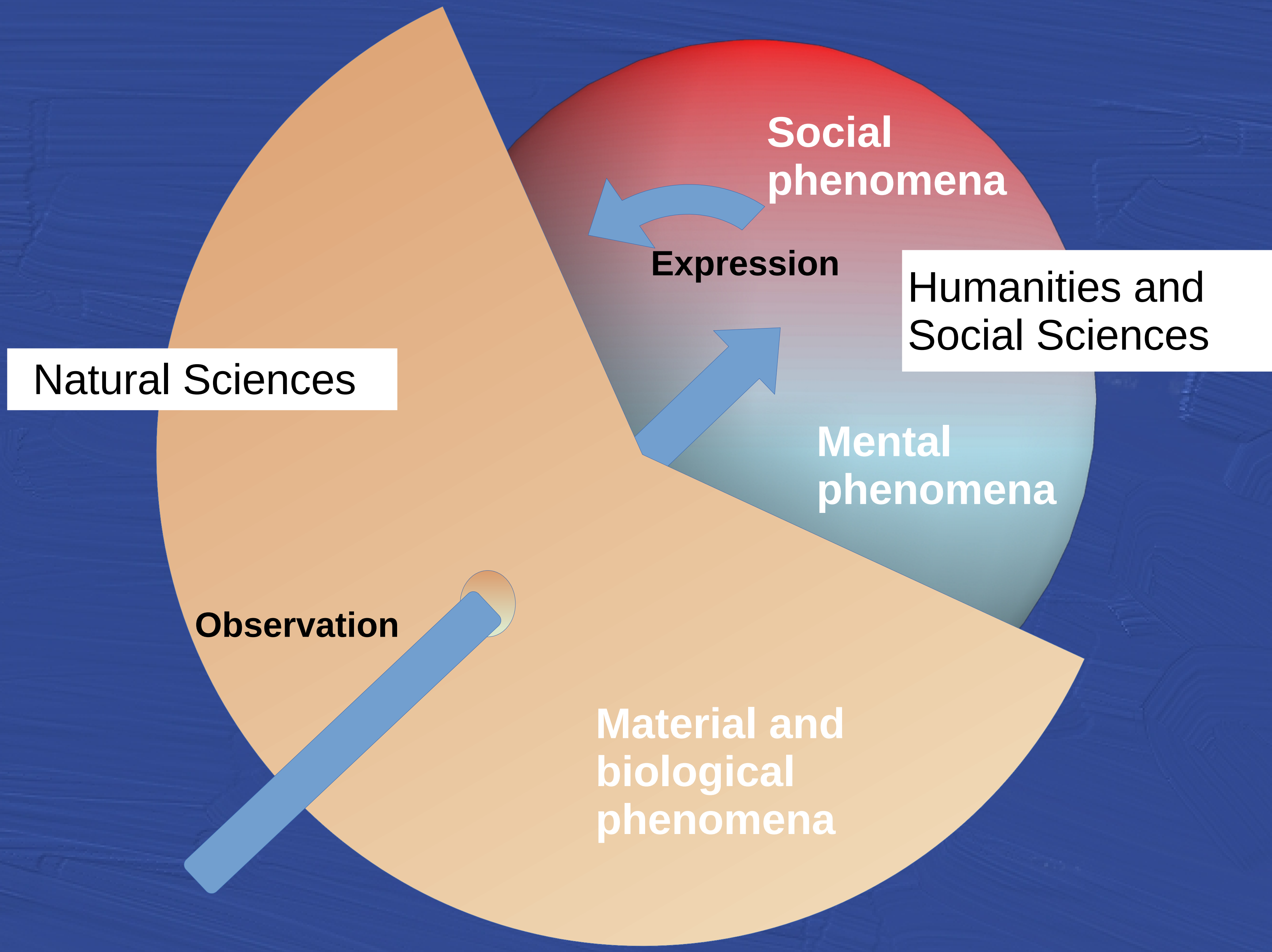
This definition of the scope or domain modelled is based on the conviction that in a constructivist approach of scientific knowledge, a conceptualisation and data model can only be developed from the point of view of a specific discipline because *scientific objects* do not exist in the absolute but depend on the method and research agenda. They depend on the perspective or epistemic context researchers adopt in considering states of affairs: *scientific objects*, and semantic models modelling them, are not declared to be the only appropriate and exclusive representation of *things* in the pre-Kantian sense but defined as *intentional objects* constructed from the point of view of a discipline and methodological approach in relation to things in the world. Scientific objects are not the things in the world themselves, even if they must necessarily refer to them by way of observation or experimentation, if a scientific and therefore realistic approach is to be maintained. This corresponds to the notion of inter-objectivity in social sciences relying on the distinction between things in themselves and things as perceived, experienced and discussed by human subjects, in their shared intentionality and in relation to their social practices and context.

sdhss.org

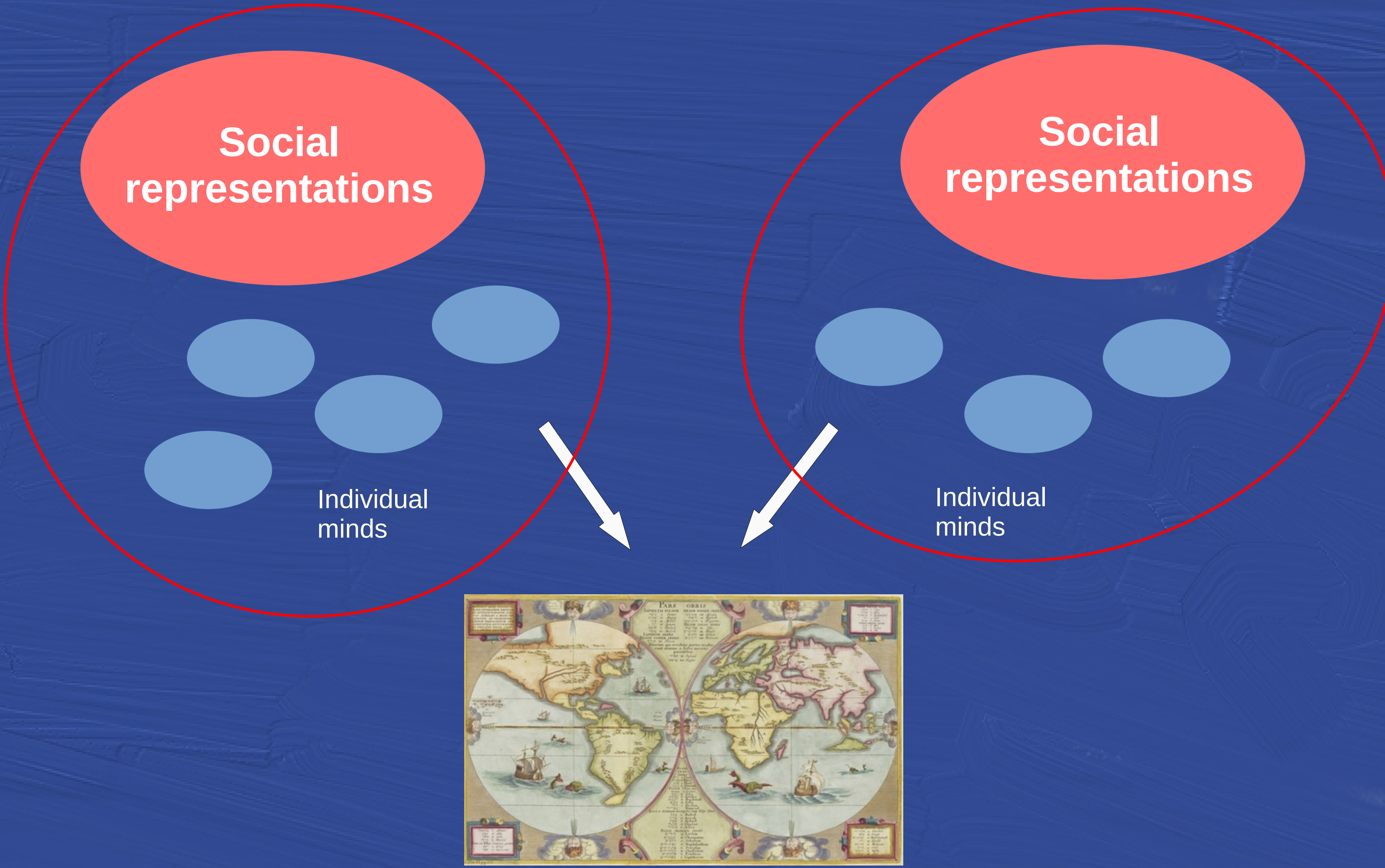


- Length of a bridge
- Color of a bicycle

- Weather in Paris in March 2024
- Economic activity of France in 2023


Social
phenomena
Expression
Humanities and
Social Sciences
Natural Sciences
Mental
phenomena
Observation
Material and
biological
phenomena

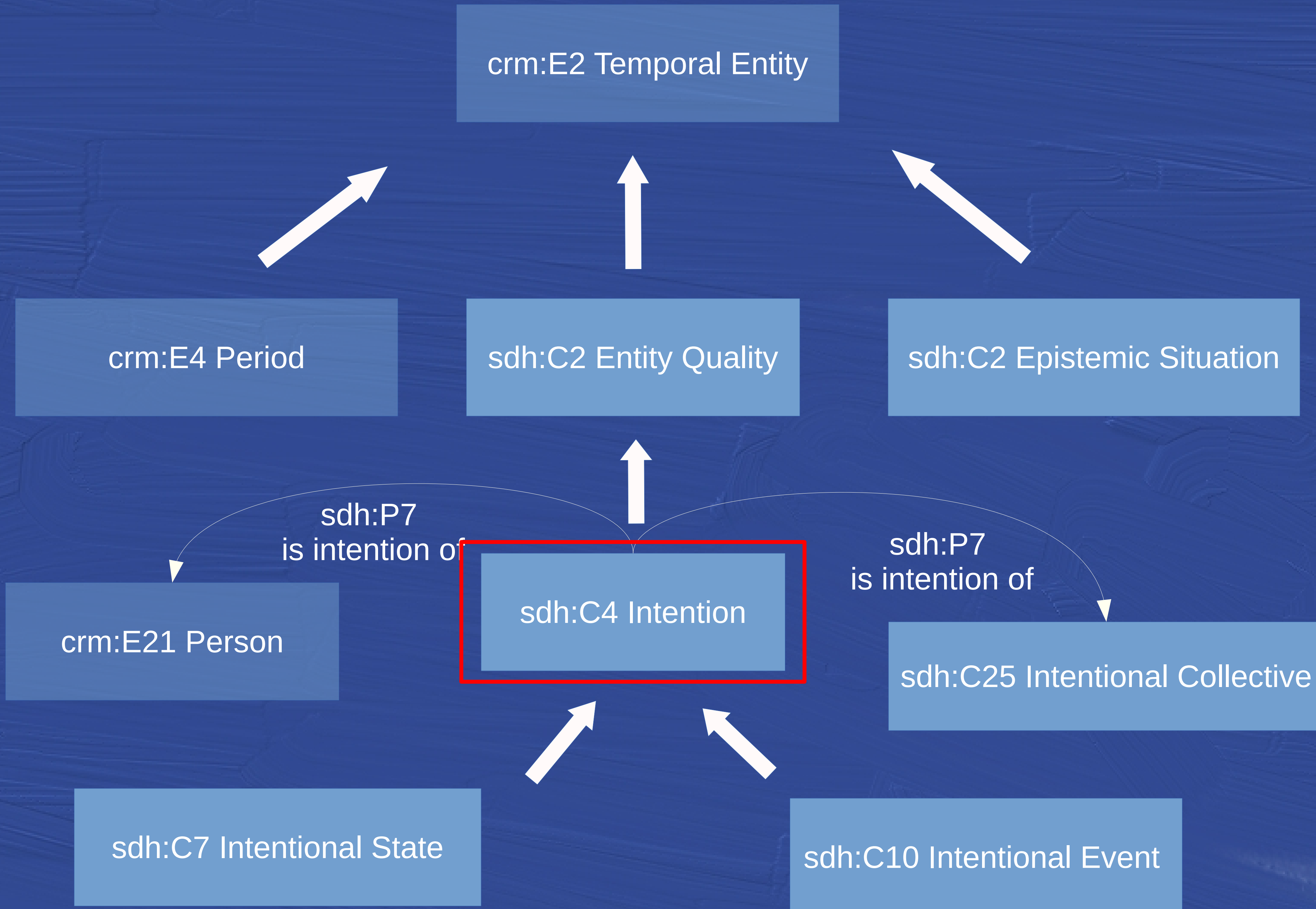
The observed phenomenon: different mental and social representations of the same phenomena



How to conceptualize observable mental and social phenomena ?

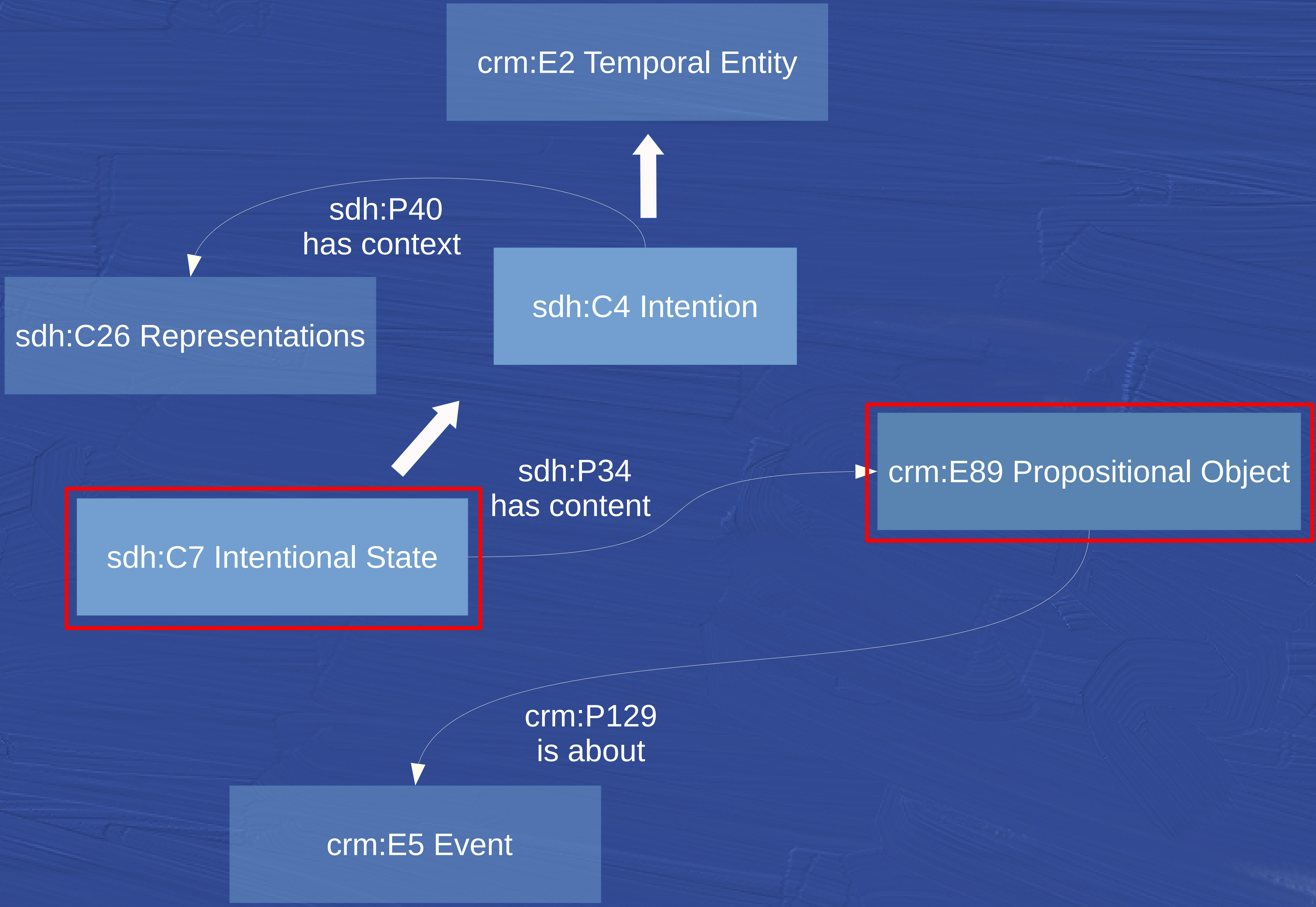
« In philosophy, **intentionality** is the power of minds and mental states to be about, to represent, or to stand for, things, properties and states of affairs. To say of an individual's mental states that they have intentionality is to say that they are mental representations or that they have contents. » https://plato.stanford.edu/entries/intentionality

« **Collective intentionality** is the power of minds to be jointly directed at objects, matters of fact, states of affairs, goals, or values. [...] Collective intentional attitudes permeate our everyday lives, for instance when two or more agents look after or raise a child, grieve the loss of a loved one, campaign for a political party, or cheer for a sports team. They are relevant for philosophers and social scientists because they play crucial roles in the constitution of the social world. https://plato.stanford.edu/entries/collective-intentionality


crm:E2 Temporal Entity
crm:E4 Period
sdh:C2 Entity Quality
sdh:C2 Epistemic Situation
sdh:P7
is intention of
crm:E21 Person
sdh:C4 Intention
sdh:P7
is intention of
sdh:C25 Intentional Collective
sdh:C7 Intentional State
sdh:C10 Intentional Event

Different perspectives on the same events: interpreting a car race as fun or as a crime

Sports car photo created by azerbaijan_stockers - www.freepik.com


crm:E2 Temporal Entity
sdh:P40
has context
sdh:C26 Representations
sdh:C4 Intention
sdh:P34
has content
crm:E89 Propositional Object
sdh:C7 Intentional State
crm:P129
is about
crm:E5 Event

Foundational ontologies
& modelling best practices

DOLCE + Descriptions and Situations
& object-oriented modelling principles

Generic, domain related core ontology

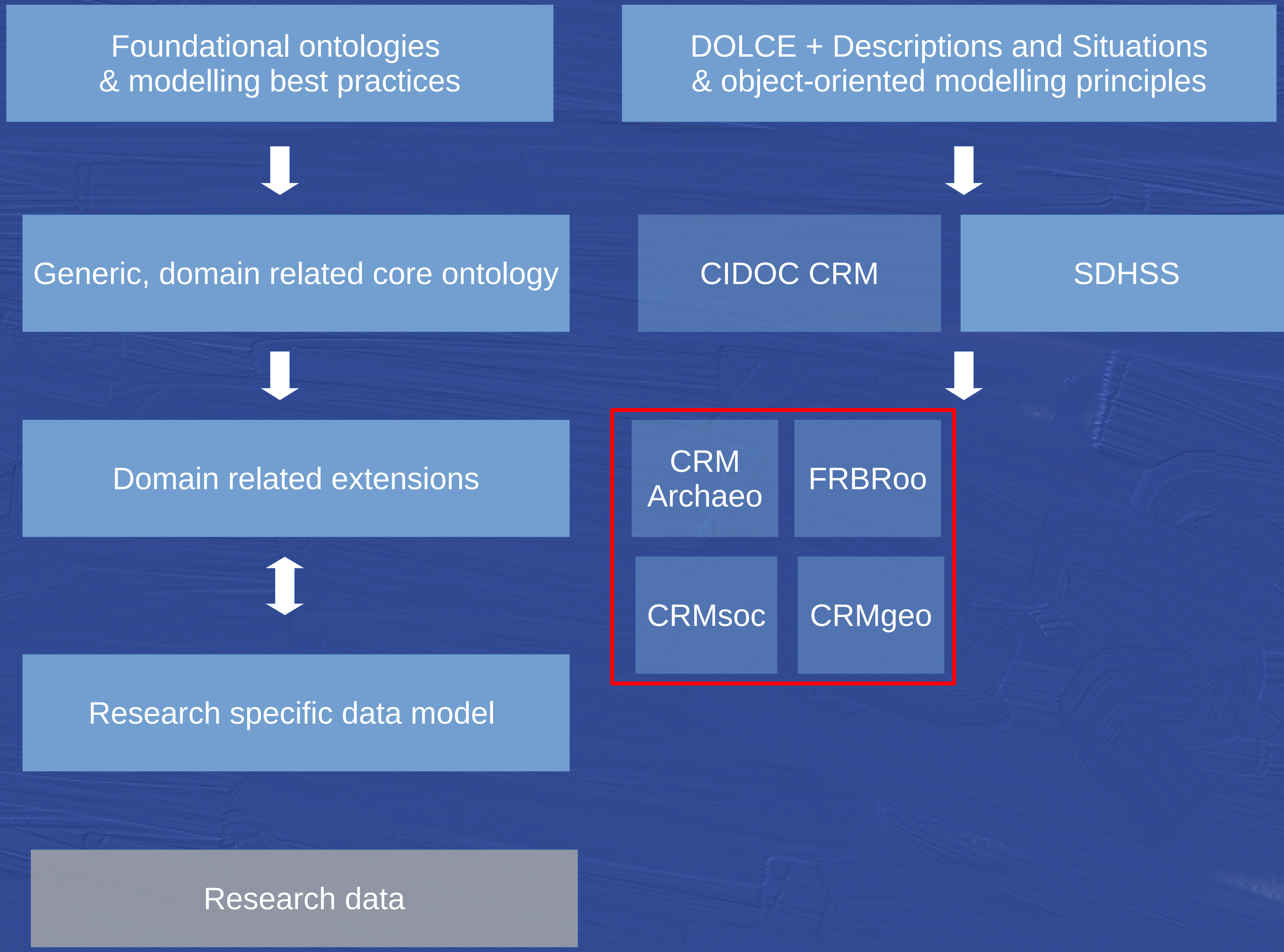
CIDOC CRM

SDHSS

Research agenda

Research specific data model

Research data


Foundational ontologies
& modelling best practices
DOLCE + Descriptions and Situations
& object-oriented modelling principles
Generic, domain related core ontology
CIDOC CRM
SDHSS
Domain related extensions
CRM Archaeo
FRBRoo
CRMsoc
CRMgeo
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CIDOC CRM

SDHSS

CRM Archaeo

FRBRoo

Society & Law (SDHSS)

Literary life (SDHSS)

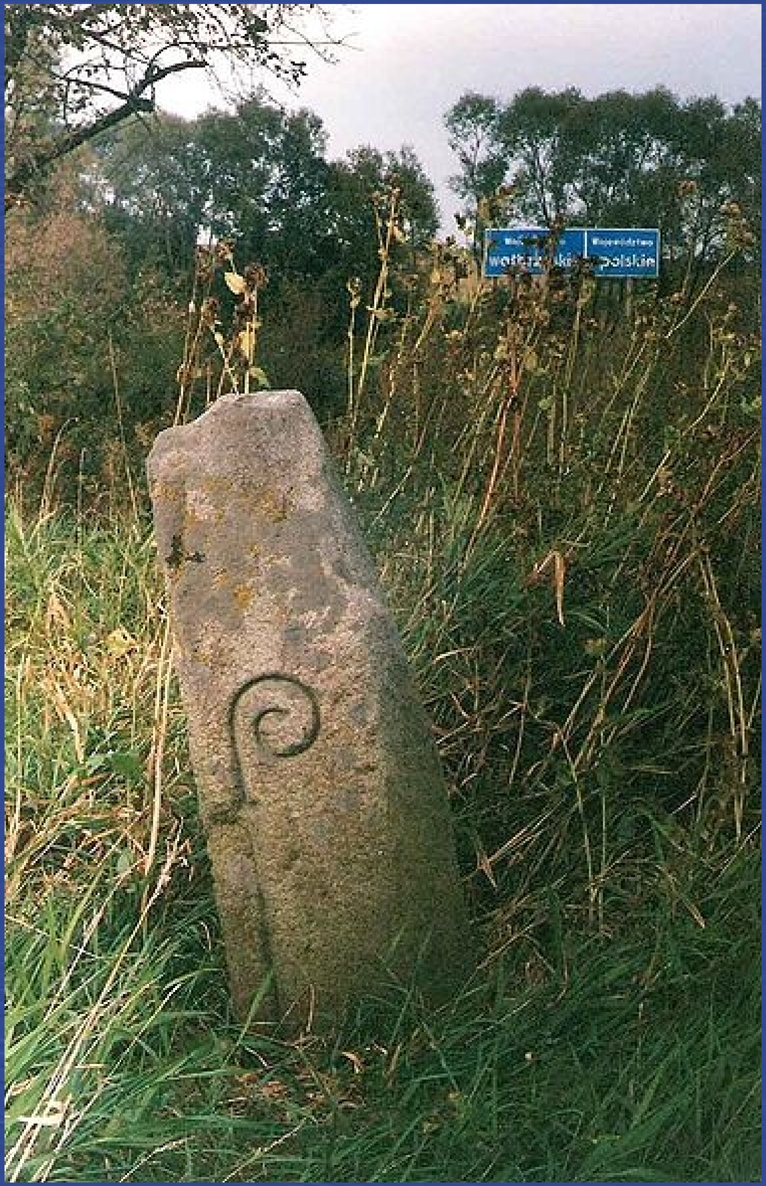
CRMsoc

CRMgeo

Education & Universities (SDHSS)

Ships & navigation (SDHSS)

Artefacts and their social significance: a bondary stone that defines a boundary



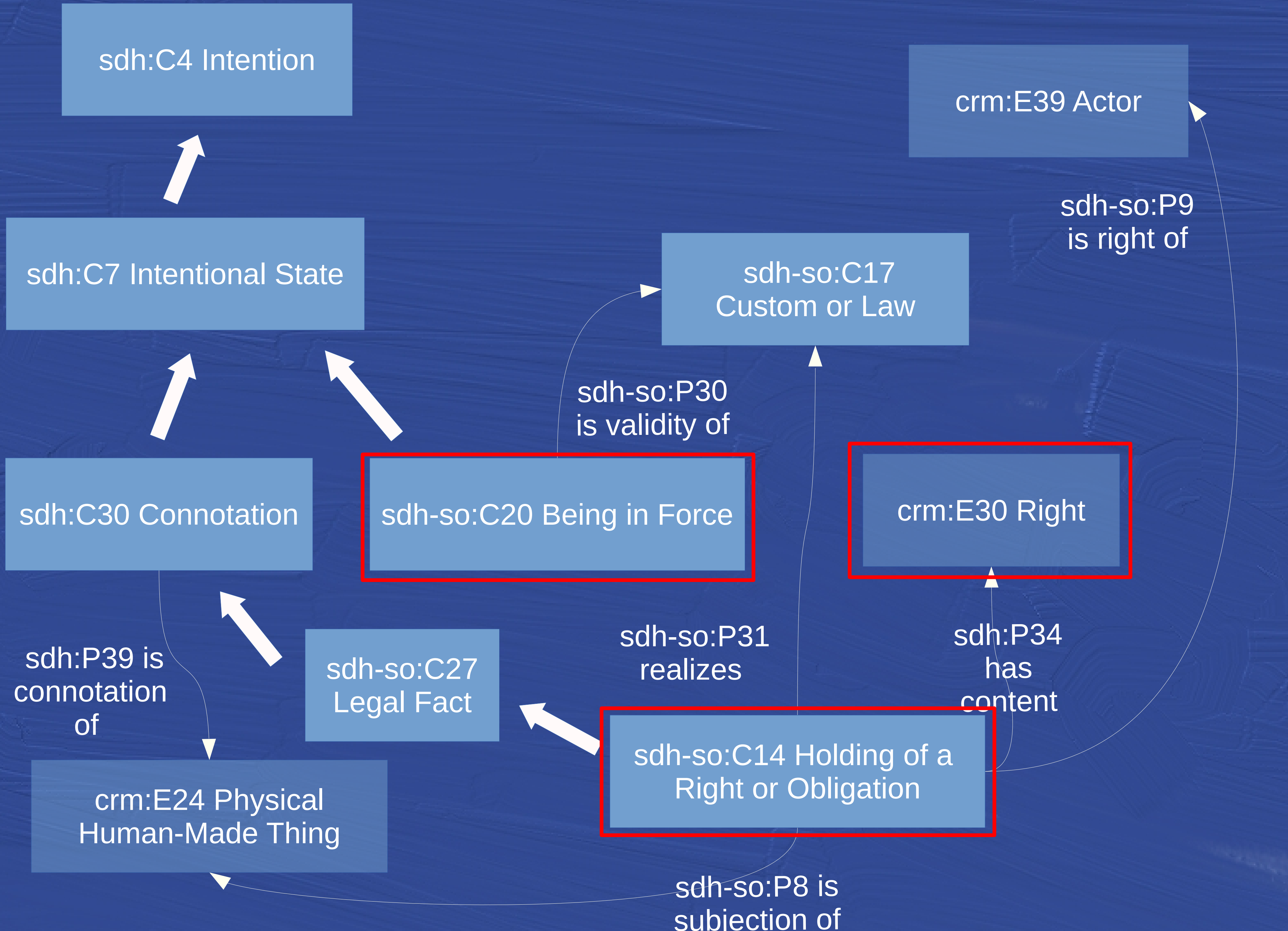
Border stone from c. 1280 on the border of the lands of the Wrocław bishops.

In the background the contemporary road sign marking the border between Opole and Wałbrzych voivodships.

Photo: Władysław Łoś (CC By SA 3.0)

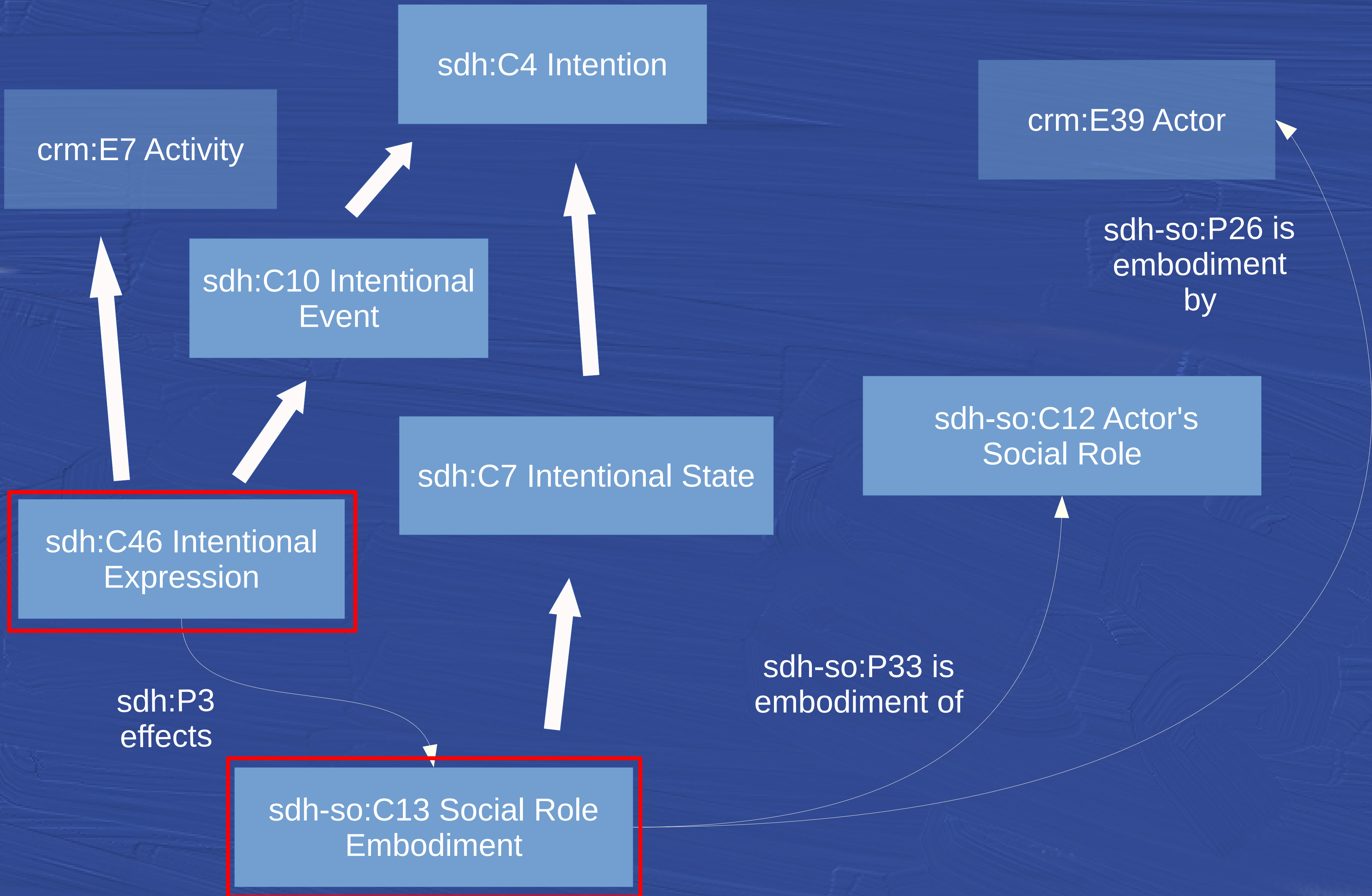
The physical object, the notion of border, the symbol of the episcopal crozier, the function of bishop.

The bio-physical world, the social world



sdh:C4 Intention
crm:E39 Actor
sdh-so:P9 is right of
sdh:C7 Intentional State
sdh-so:C17 Custom or Law
sdh-so:P30 is validity of
sdh:C30 Connotation
sdh-so:C20 Being in Force
crm:E30 Right
sdh:P39 is connotation of
sdh-so:C27 Legal Fact
sdh-so:P31 realizes
sdh:P34 has content
sdh-so:C14 Holding of a Right or Obligation
crm:E24 Physical Human-Made Thing
sdh-so:P8 is subjection of

Social roles: the rite that establishes the function


sdh:C4 Intention
crm:E7 Activity
crm:E39 Actor
sdh-so:P26 is embodiment by
sdh:C10 Intentional Event
sdh-so:C12 Actor's Social Role
sdh:C7 Intentional State
sdh:C46 Intentional Expression
sdh-so:P33 is embodiment of
sdh:P3 effects
sdh-so:C13 Social Role Embodiment

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Research specific data model

sdhss.org

Research data

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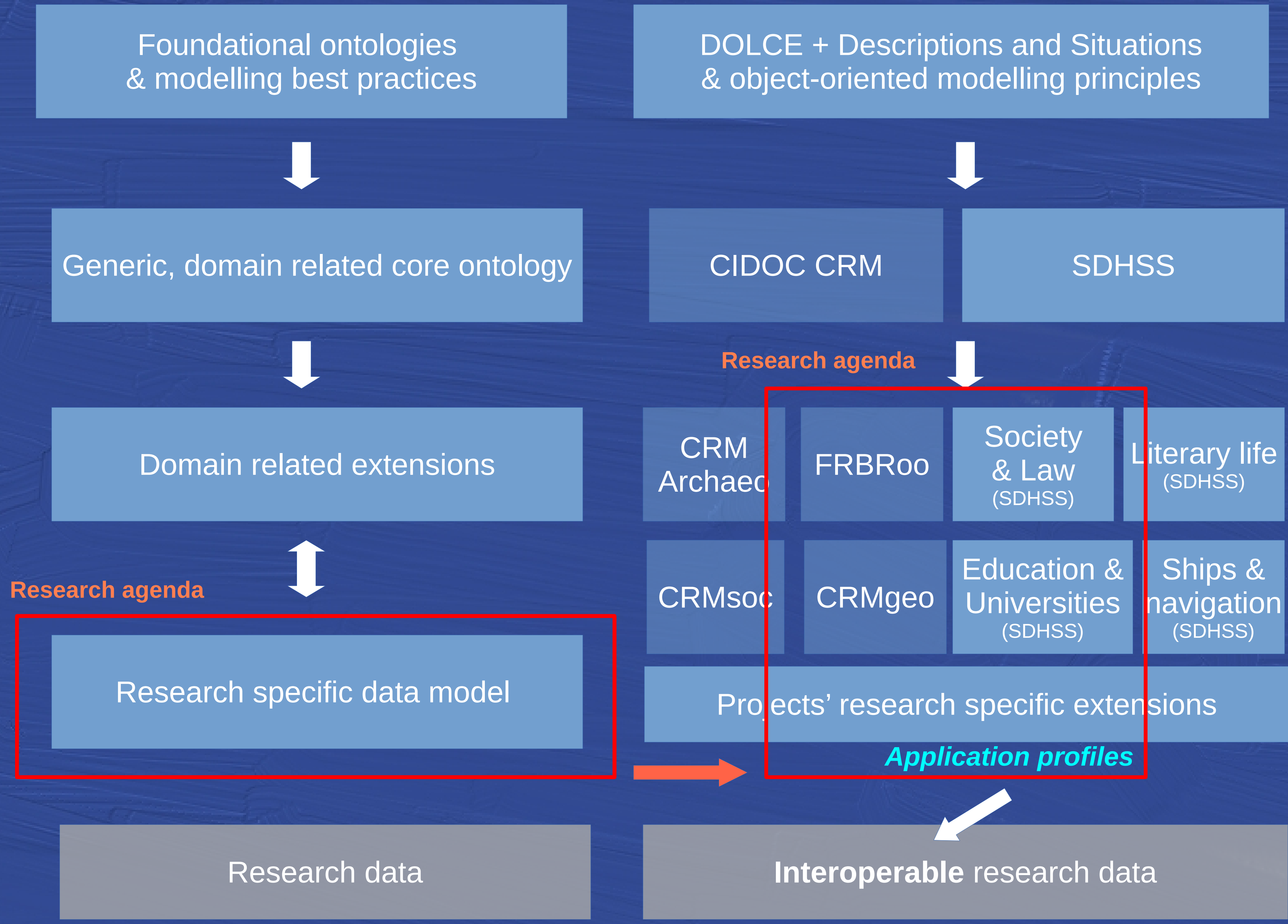
Ships & navigation (SDHSS)

Projects' research specific extensions


Foundational ontologies
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Generic, domain related core ontology
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Ships & navigation (SDHSS)
Research specific data model
Projects' research specific extensions
Application profiles
Research data
Interoperable research data

The SDHSS ontology ecosystem

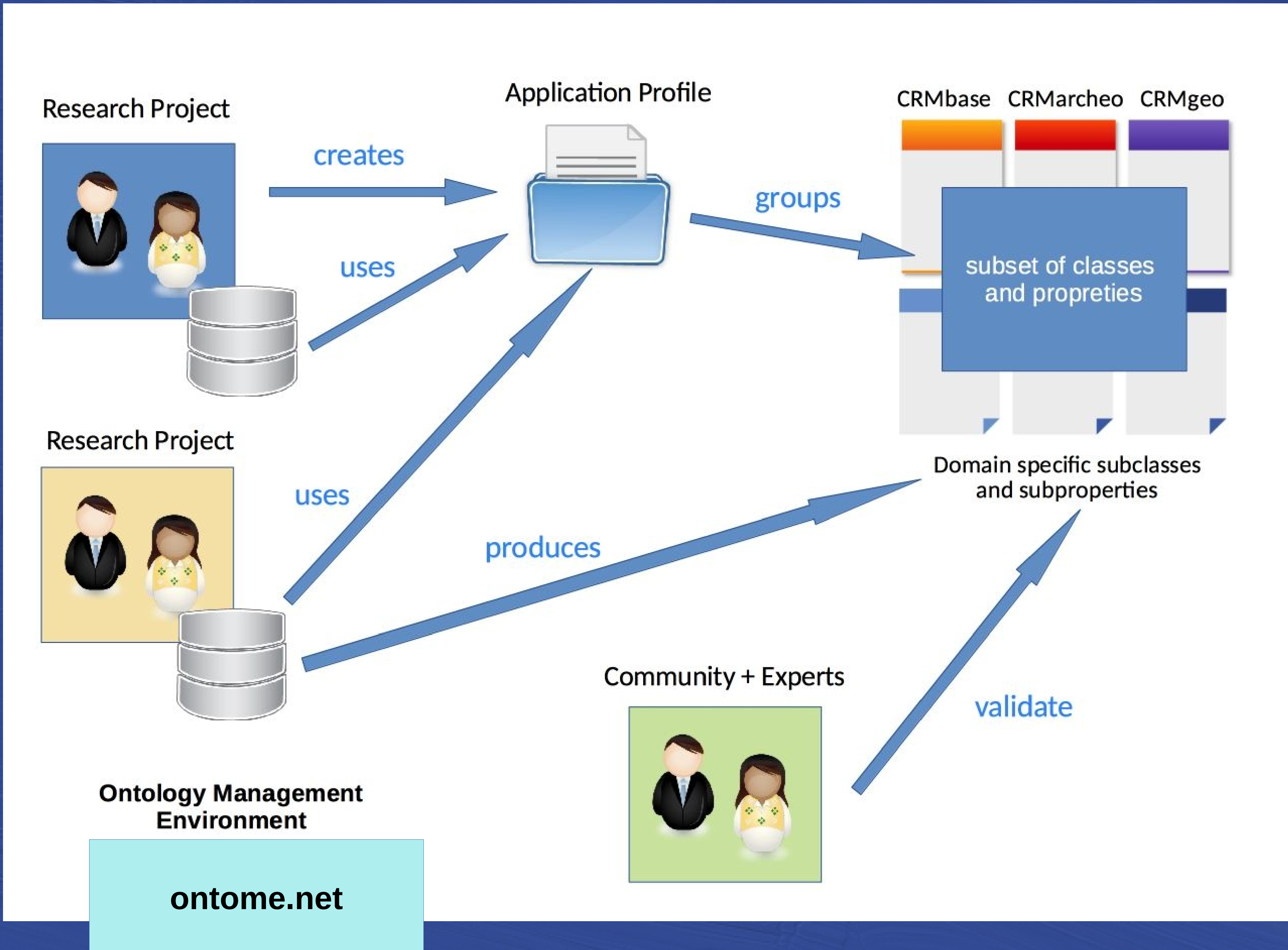
A proposal to address the challenges posed by the FAIR principles in the HSS domain :

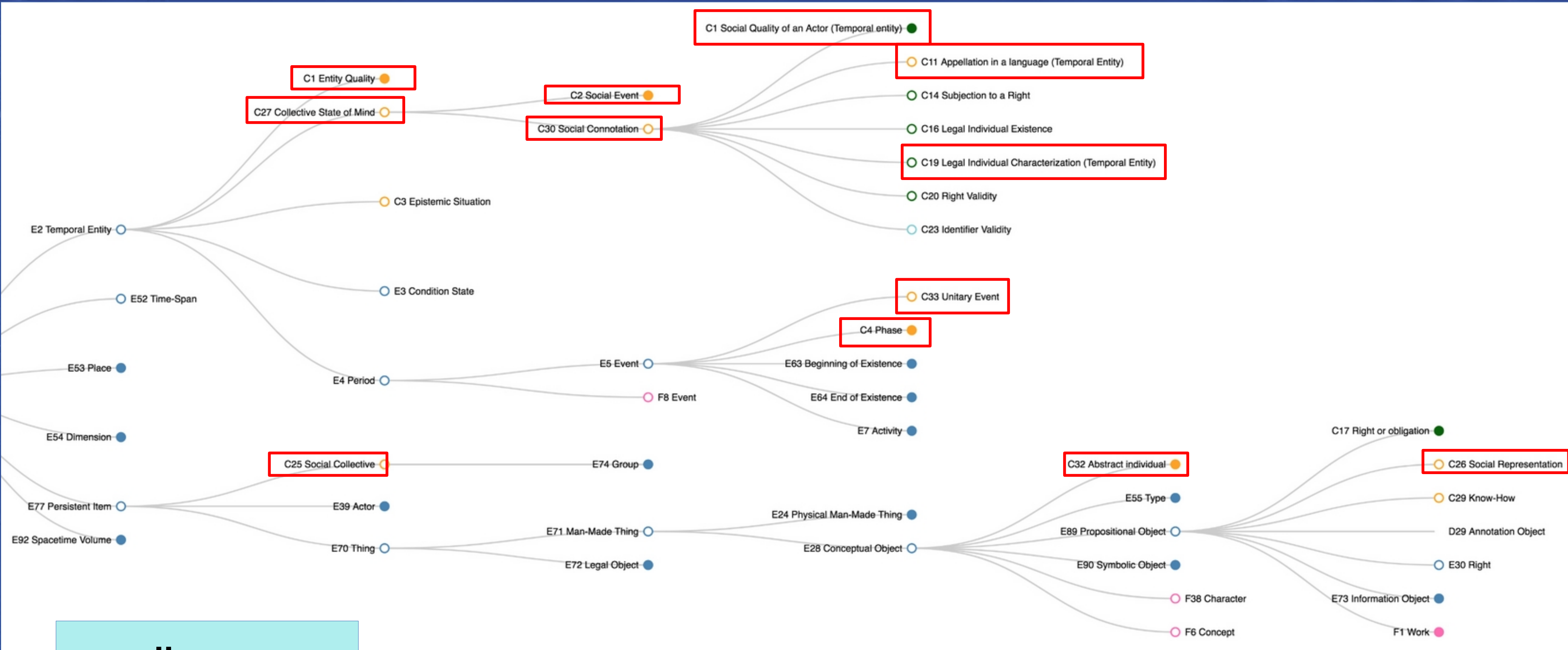
I1. (meta)data use a *formal, accessible, shared, and broadly applicable* ***language*** *for* ***knowledge representation***.

R1.3. (meta)data meet ***domain-relevant community standards***.

The challenge of infrastructure and community building

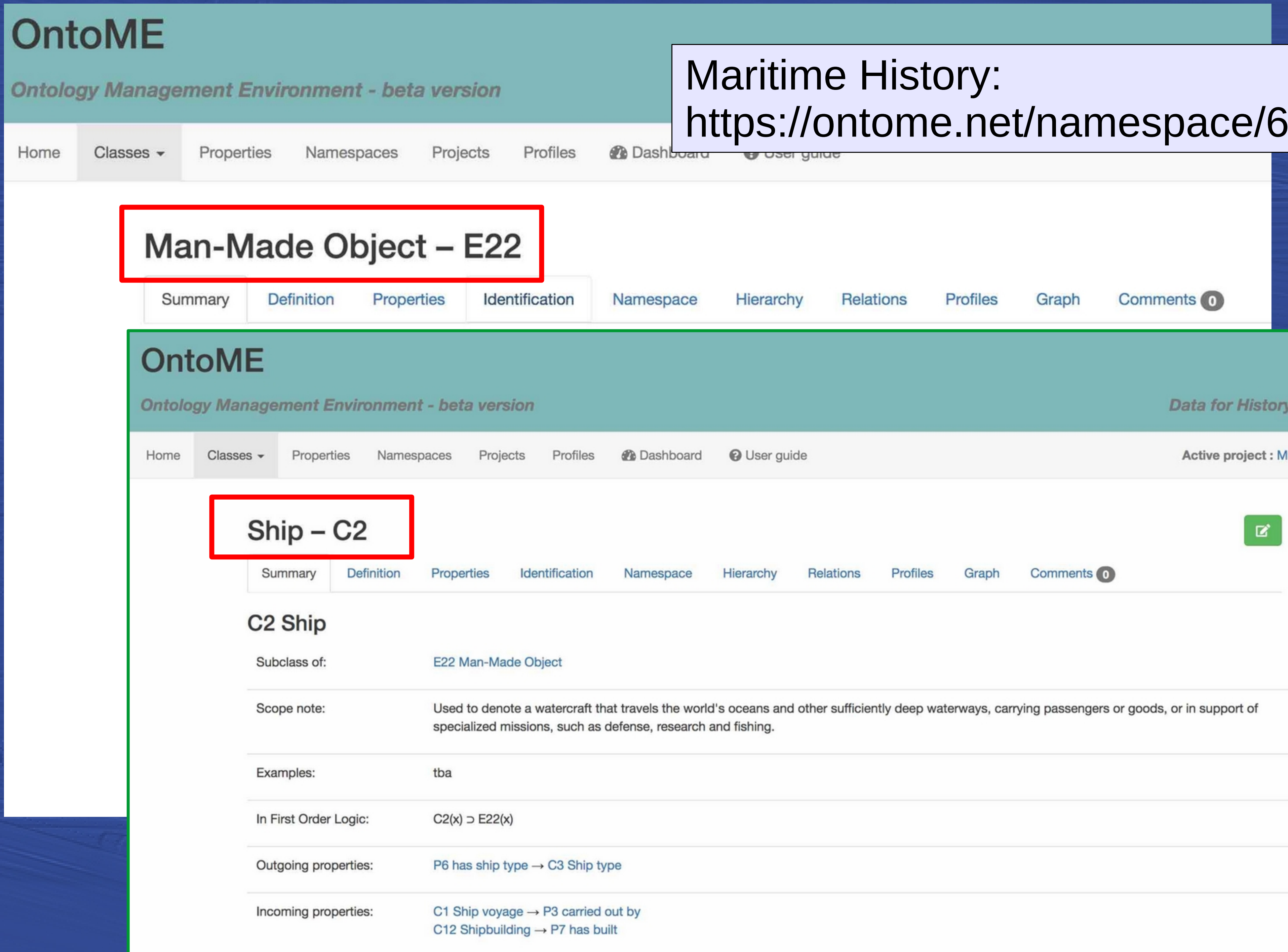
Aspects of the LOD4HSS initiative

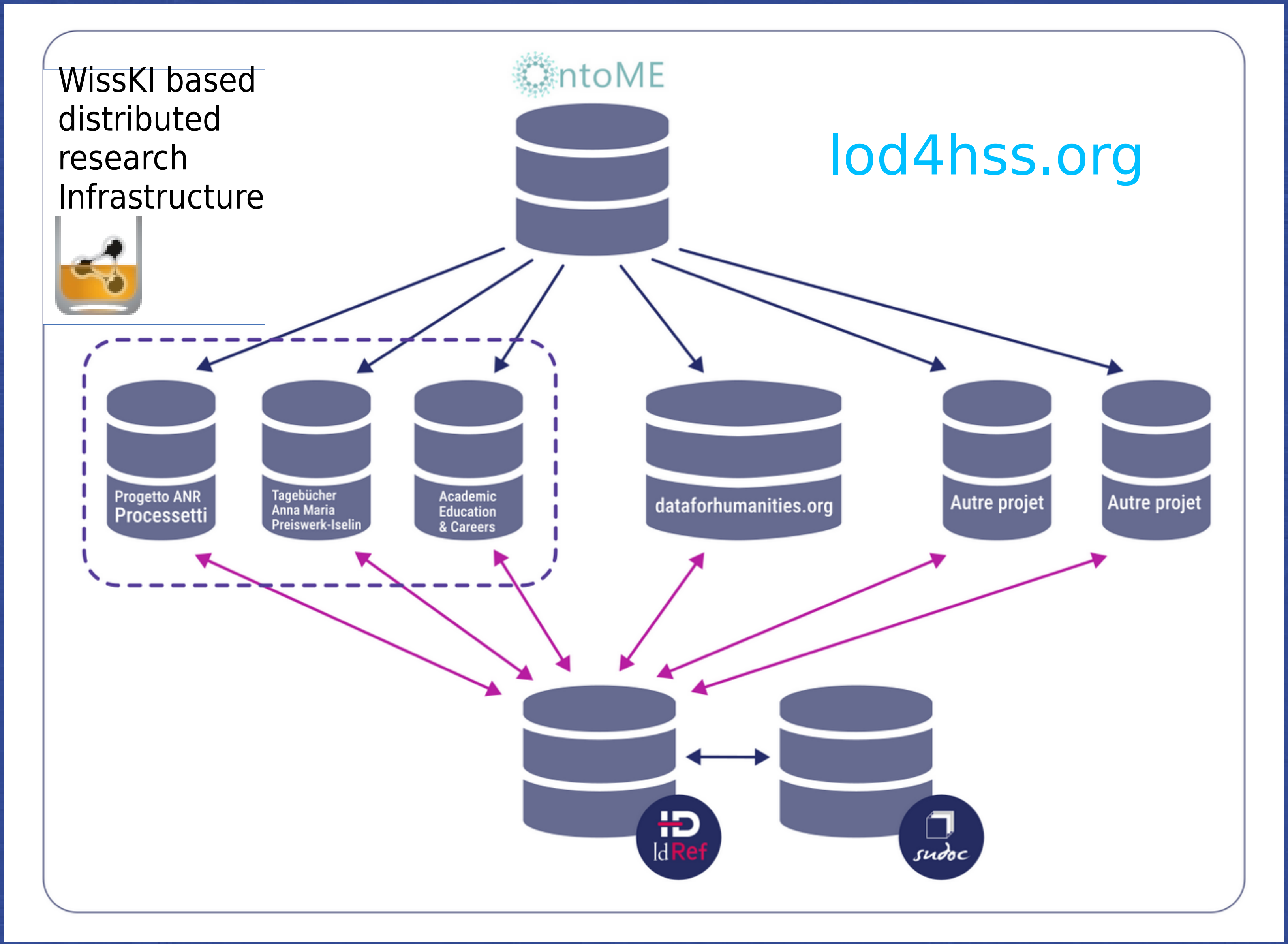

Research Project
creates
uses
Application Profile
groups
CRMbase
CRMarcheo
CRMgeo
subset of classes
and propreties
Domain specific subclasses
and subproperties
Research Project
uses
produces
Community + Experts
validate
Ontology Management
Environment
ontome.net


C1 Social Quality of an Actor (Temporal entity)
C11 Appellation in a language (Temporal Entity)
C1 Entity Quality
C2 Social Event
C27 Collective State of Mind
C30 Social Connotation
C14 Subjection to a Right
C16 Legal Individual Existence
C19 Legal Individual Characterization (Temporal Entity)
C3 Epistemic Situation
C20 Right Validity
E2 Temporal Entity
C23 Identifier Validity
E3 Condition State
E52 Time-Span
C33 Unitary Event
C4 Phase
E53 Place
E5 Event
E63 Beginning of Existence
E4 Period
E64 End of Existence
F8 Event
E7 Activity
E54 Dimension
C17 Right or obligation
C25 Social Collective
E74 Group
C32 Abstract individual
C26 Social Representation
E55 Type
C29 Know-How
E77 Persistent Item
E39 Actor
E24 Physical Man-Made Thing
E71 Man-Made Thing
E89 Propositional Object
D29 Annotation Object
E92 Spacetime Volume
E70 Thing
E28 Conceptual Object
E72 Legal Object
E90 Symbolic Object
E30 Right
F38 Character
E73 Information Object
F6 Concept
F1 Work

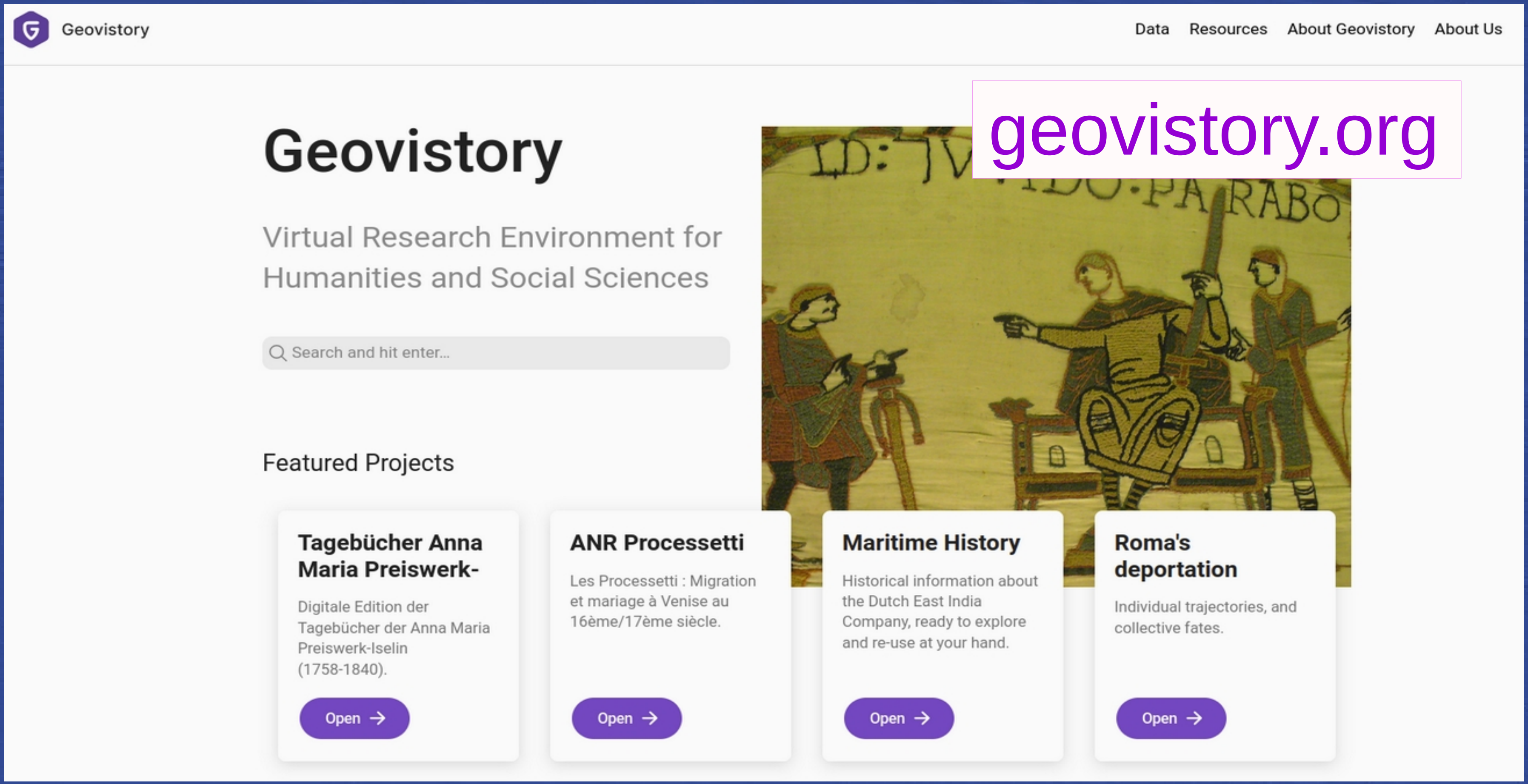
sdhss.org

Maritime History:
https://ontome.net/namespace/66





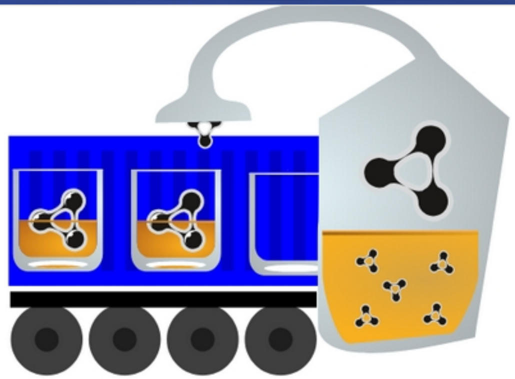
LOD4HSS as a community driven, distributed research infrastructure



Geovistory (2019-2025) : a LOD repository and virtual research environment for storing and sharing open research data (geovistory.org)

WissKI Distillery

For more information, see https://github.com/FAU-CDI/wisski-distillery.



WissKIs on this Distillery

fberetta

https://fberetta.lod4hss.cloud/

humility

https://humility.lod4hss.cloud/

knurg

https://knurg.lod4hss.cloud/

nepoc

https://nepoc.lod4hss.cloud/

sandbox

https://sandbox.lod4hss.cloud/

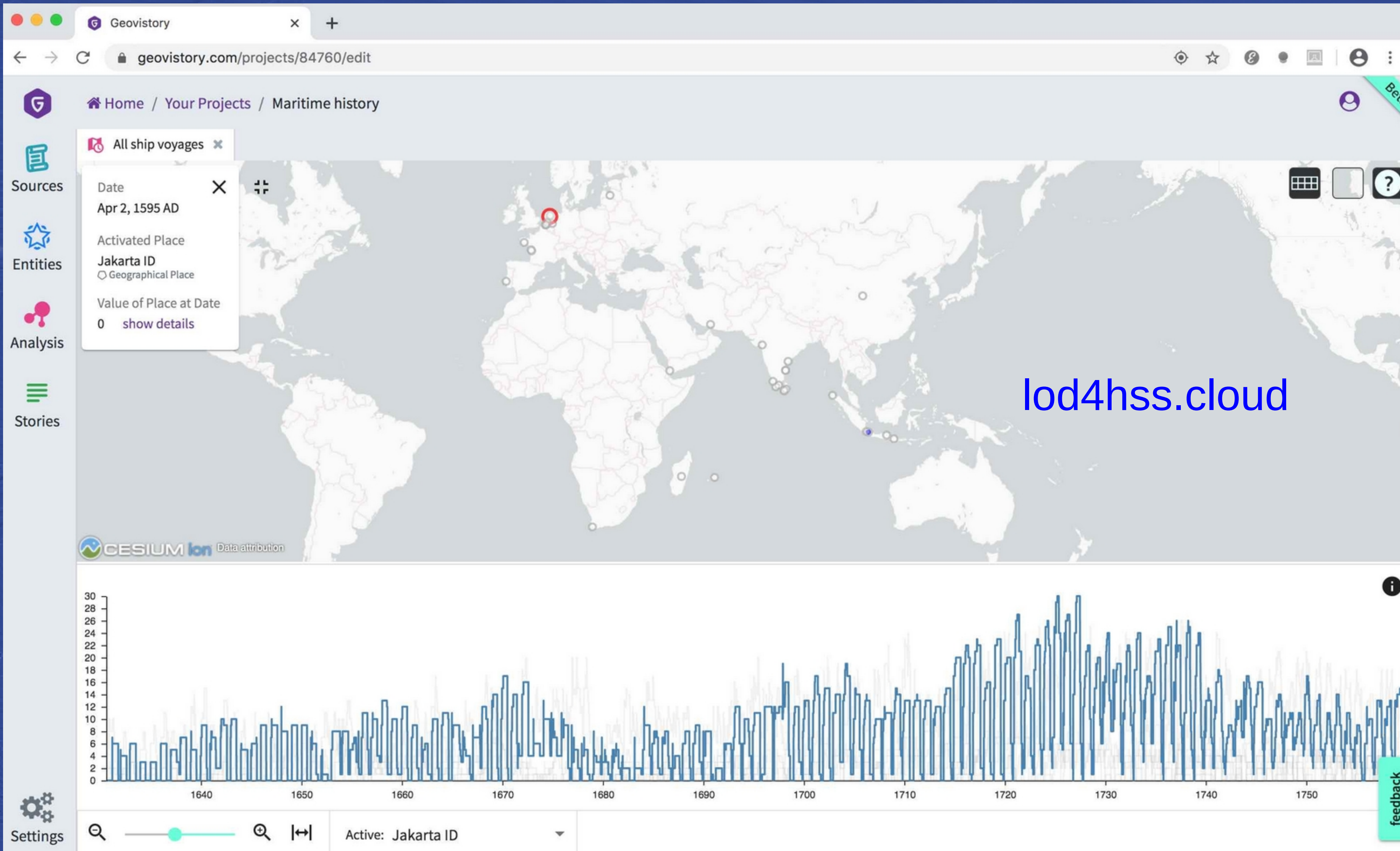
sandbox2

https://sandbox2.lod4hss.cloud/

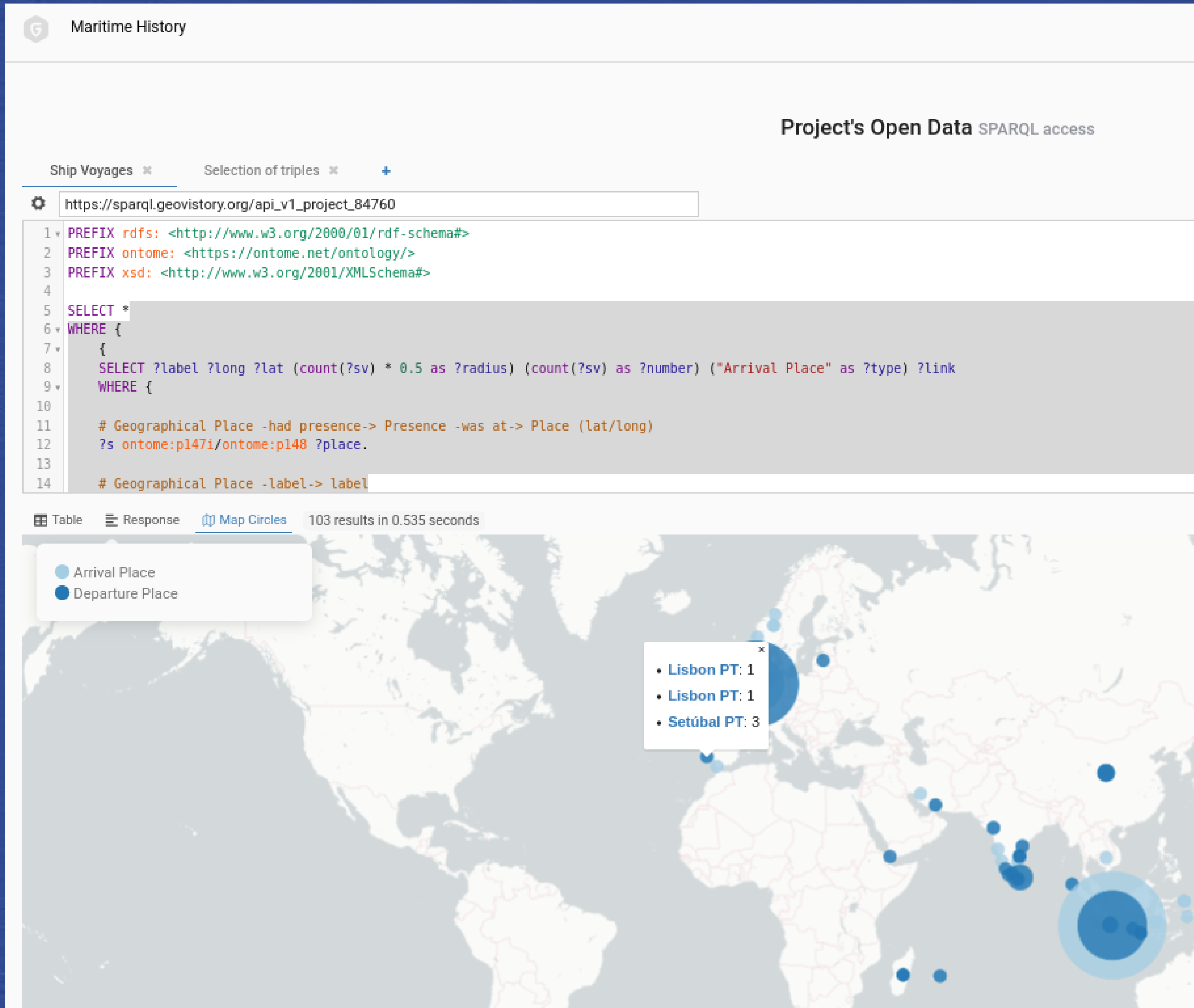
swiss-elites

https://swiss-elites.lod4hss.cloud/

lod4hss.cloud



Virtual research environments for HSS research …



… open and connected to the Semantic Web.

LOGRE (LOcal GRaph Editor)

LOD4HSS Home Projects Data Guides About us

LOD4HSS Initiative

For the semantisation and reuse of research data in HSS

[Please note that this website is currently undergoing a major redesign]

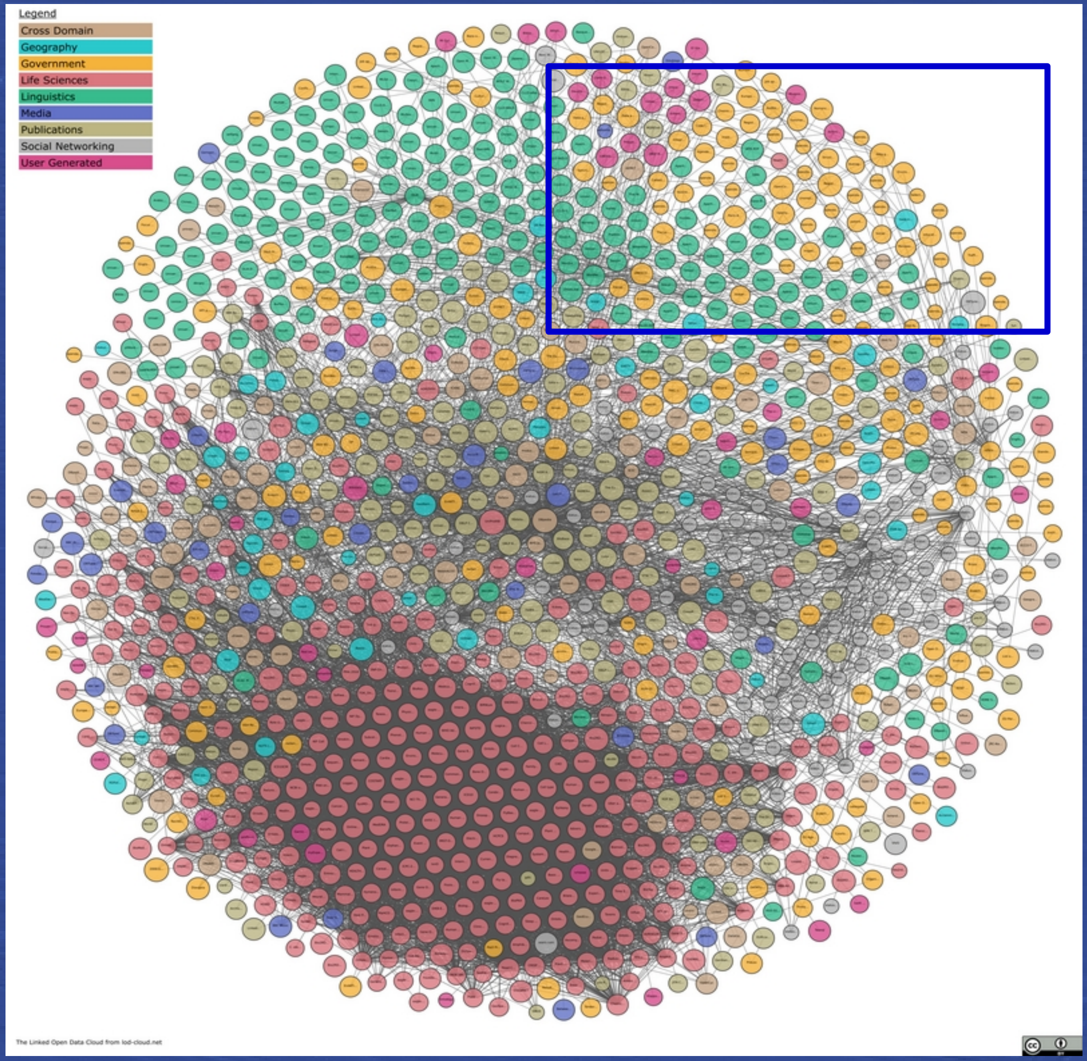
The LOD4HSS initiative, promoted by the Laboratoire de Recherche Historique Rhône-Alpes (LARHRA, Lyon) and the Digital Humanities Chair at the University of Bern, aims to establish a sustainable and vibrant community that manages an ecosystem of projects and infrastructures for research in the Humanities and Social Sciences based on semantic web technologies. Open to all interested institutions and projects, the initiative dates back to 2017, with the creation of the OntoME web platform (ontome.net) and the launch of the Semantic Data for the Humanities and Social Sciences (SDHSS) project (sdhss.org).

The LOD4HSS initiative is built around four main pillars:

1. **A common ontology ecosystem.** At the heart of this endeavour is the SDHSS project. This project provides a methodology for ontology engineering based on OntoClean and DOLCE, as well as a common semantic backbone based on CIDOC CRM, a core ontology for describing events, actors, sources and other relevant information in the HSS domain. The SDHSS project meets the basic needs of most projects and provides a set of best practices for creating community-driven extensions for research subdomains. This ensures semantic consistency and interoperability across different datasets, enabling the construction of a research-driven, distributed giant knowledge graph.

lod4hss.org

Open and reusable HSS research open data as a component of the Semantic Web



Some final thoughts

Can we achieve these goals in HSS research ?

- **Transformative change** concerns the transfer of analogue information and practices to a digital form
- **Enabling change** is the use of data-intensive technologies to *address [new] research questions* that could not be tackled in another form
- ***Substitutive change***, digital technologies are used to support or even replace conceptual parts of the research process

DFG, White Paper 2020

We have all the tools we need to undertake the digital transformation in HSS...

We have all the tools we need to undertake the digital transformation in HSS...

... but we're not quite there yet!

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... but we're not quite there yet!

We need to:

- train students in digital skills from their undergraduate studies onwards, and provide continuous training;

We have all the tools we need to undertake the digital transformation in HSS...

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We need to:

- train students in digital skills from their undergraduate studies onwards, and provide continuous training;
- convince the authorities to fund sustainable, large-scale infrastructure;

We have all the tools we need to undertake the digital transformation in HSS...

... but we're not quite there yet!

We need to:

- train students in digital skills from their undergraduate studies onwards, and provide continuous training;
- convince the authorities to fund sustainable, large-scale infrastructure;
- build disciplinary communities that develop new digital research approaches and agendas based on open data and digital methods, and driven by a collaborative research approach.

Join the project and make HSS research digital!

sdhss.org

lod4hss.org

lod4hss.cloud

LOD4HSS
Linked Open Data for the Humanities & Social Sciences

